



Math Klt JCE 250603 094007

agricultural science (Major Tutions Botswana-Academy)



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BOTSWANA EXAMINATIONS COUNCIL
JUNIOR CERTIFICATE EXAMINATION

MATHEMATICS

13/2

Paper 2

October/November 2018

Marks: 100

Time: 2 Hours

Candidate

Full Names:

Centre Number:

J	C				
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Candidate Number:

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INSTRUCTIONS

1. Write your examination number in the space provided above.
2. Answer **ALL** questions.
3. **ALL** answers must be written in the spaces provided.
4. Show all the necessary working.
5. Calculators may be used in this paper.
6. Use the value π in your calculator unless otherwise stated.
7. If the degree of accuracy is **not** specified in the question and the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

FOR EXAMINER'S USE ONLY

Section	Marks Scored
A	
B	
C	
Total Marks	

This question paper contains 20 printed pages.

SECTION A

(30 Marks)

Answer **all** questions in this section.

1. Write down the two prime numbers between 50 and 60.

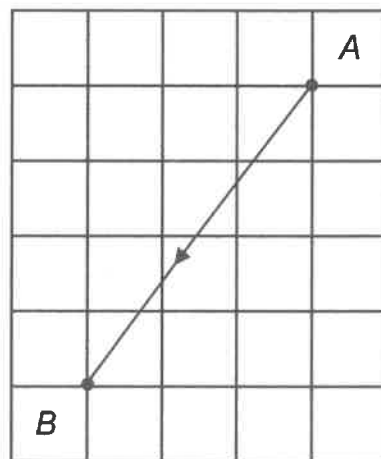
....., (2)

2. A box contains sweets of the same size, shape and texture. However, there are 3 different flavours. There are 2 chocolate, 3 mint and 7 strawberry flavoured sweets in the box.

Calculate the probability of randomly picking a strawberry flavoured sweet from the box.

..... (2)

3. Vector AB is shown on the grid below.



Write down $\vec{AB}$ as a column vector.

$$\vec{AB} = \begin{pmatrix} \\ \end{pmatrix} \quad (1)$$

4. Solve $5(x - 4) = 3x - 7$.

$x = \dots\dots\dots$ (3)

5. The value of a car depreciated from P90 000 in June 2015 to P76 500 in June 2016.
Calculate the percentage depreciation of the value of the car between June 2015 and June 2016.

$\dots\dots\dots$ (2)

6. The height of a triangle is 16 cm and its base is 9.2 cm. The height of a similar triangle is 12 cm.

Calculate the base of the similar triangle.

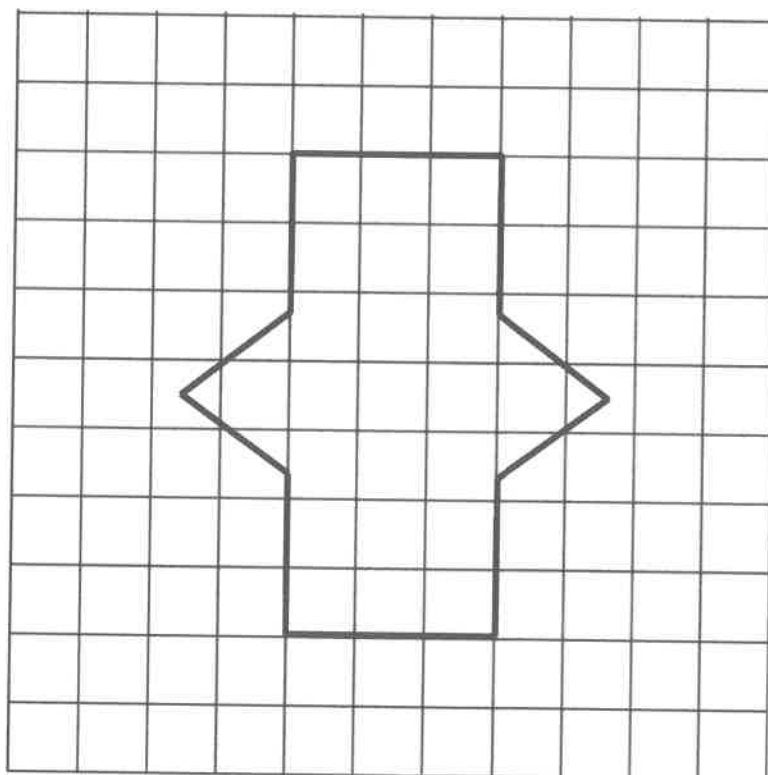
$\dots\dots\dots$ (2)

7. A 4 m ladder leans against a vertical wall at an angle of 50° from a horizontal ground. One end of the ladder rests on the top edge of the vertical wall.

What is the height of the vertical wall?

..... (2)

8. Draw all the lines of symmetry of the polygon shown below.



(2)

9. A circle has an area of 59.45 cm^2 .

Calculate the radius of the circle.

Take π as 3.142.

..... (3)

10. Solve the following simultaneous equations.

$$x + 2y = 9$$

$$y = 3x + 1$$

$x = \dots\dots\dots$, $y = \dots\dots\dots$ (3)

11. A sequence of numbers is given as;

-3, 1, 5, 9, 13, ...

- (a) Write down the next two terms of the sequence.

..... (2)

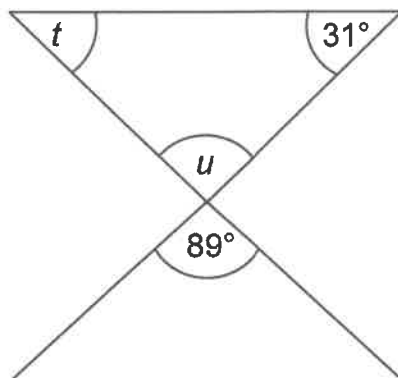
- (b) What is the common difference for the sequence?

..... (1)

- (c) Write down the general formula, in terms of n , for the sequence.

..... (1)

In the diagram below, the size of two angles is given and the size of the other two angles is marked as t and u . Use the diagram to answer question 12.



12. (a) State, with a reason, the size of the angle marked u .

.....
 (2)

- (b) Calculate the size of the angle marked t .

..... (2)



SECTION B

(50 Marks)

Answer **all** questions in this section.

13. A car started a journey at 11:20 am and completed it at 2:35 pm.

(a) Write down 2:35 pm using 24 hour clock notation.

..... (1)

(b) Calculate the time, in hours and minutes, that the car took to complete the journey.

.....hours.....minutes. (2)

(c) The total distance covered during the journey was 234 km.

Calculate the average speed, in km/h, of the car.

..... km/h (2)

14. A farmer had 250 chickens to sell at a market. The cost of one chicken was P48.

(a) He sold 8% of the chickens in the morning.

Calculate the number of chickens that the farmer sold in the morning.

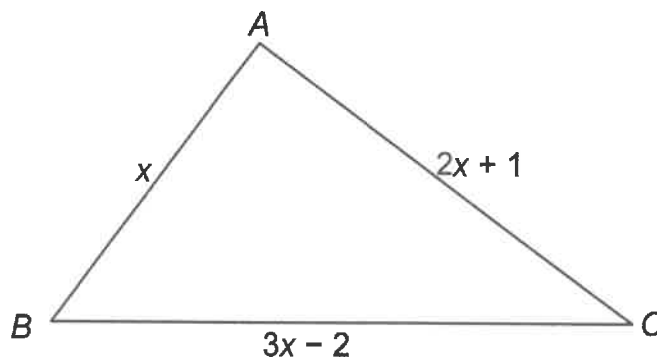
..... (2)

(b) The farmer then sold 36 chickens in the afternoon.

Calculate the total amount of money made from the sale of the chickens in the morning and in the afternoon.

..... (2)

15. Triangle ABC is shown below.



(a) Form an expression, in terms of x , for the perimeter of triangle ABC .

..... (1)

(b) The perimeter of the triangle is 35 cm.

Form an equation, in terms of x , for the perimeter of the triangle.

..... (1)

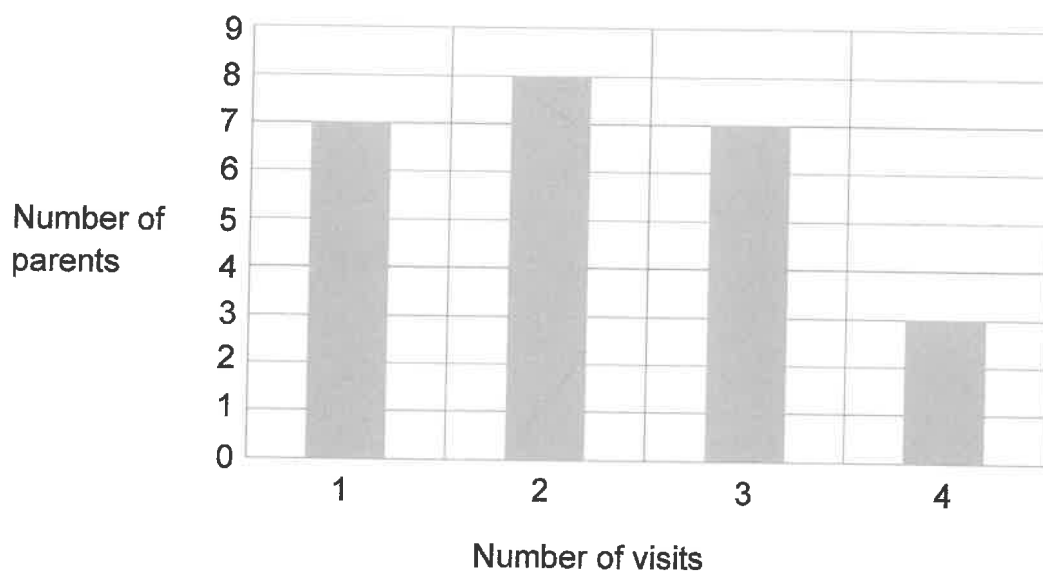
(c) Solve the equation in 15 (b).

..... (2)

(d) Calculate the length of side AC .

..... (1)

16. The bar chart below shows the number of visits made by 25 parents to a school.



- (a) State the modal number of visits.

..... (1)

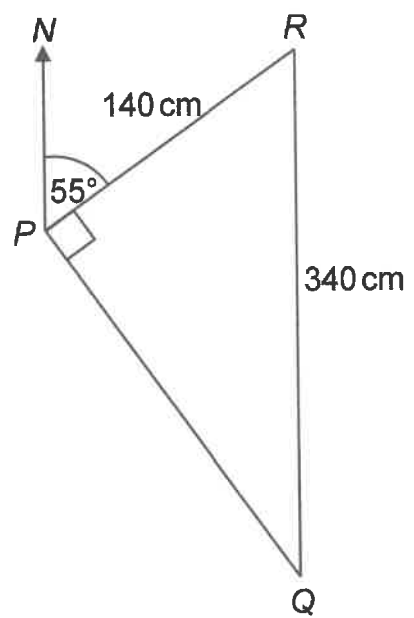
- (b) Calculate the total number of visits by the parents.

..... (2)

- (c) Calculate the mean number of visits by the parents.

..... (2)

17. The diagram below shows the positions of three points P , Q and R .



(a) What is the bearing of point Q from point P ?
..... (1)

(b) Calculate the size of angle PRQ .
..... (2)

(c) Calculate the distance between point P and point Q .
..... (2)



18. (a) Factorise $x^2 + 9x + 20$ completely.

..... (2)

- (b) Solve the equation $x^2 + 9x + 20 = 0$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ (2)

19. Labour charge, in Pula, for repairing any machine for h hours is given as $L = 300 + 120h$. The labour charge for repairing a photocopying machine was P1056.

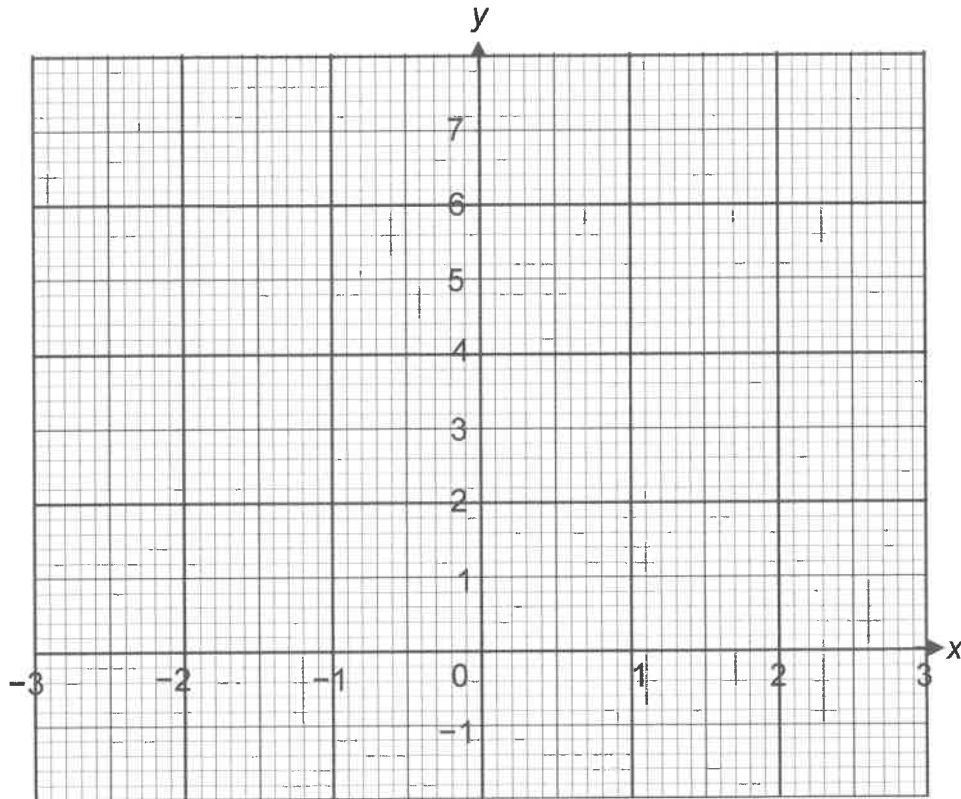
How many hours were spent on repairing the machine?

..... (3)

20. The table below shows values of x and corresponding values of y for the equation $y = x^2 - x$.

x	-2	-1	0	1	2	3
y	6	2	0	0	2	6

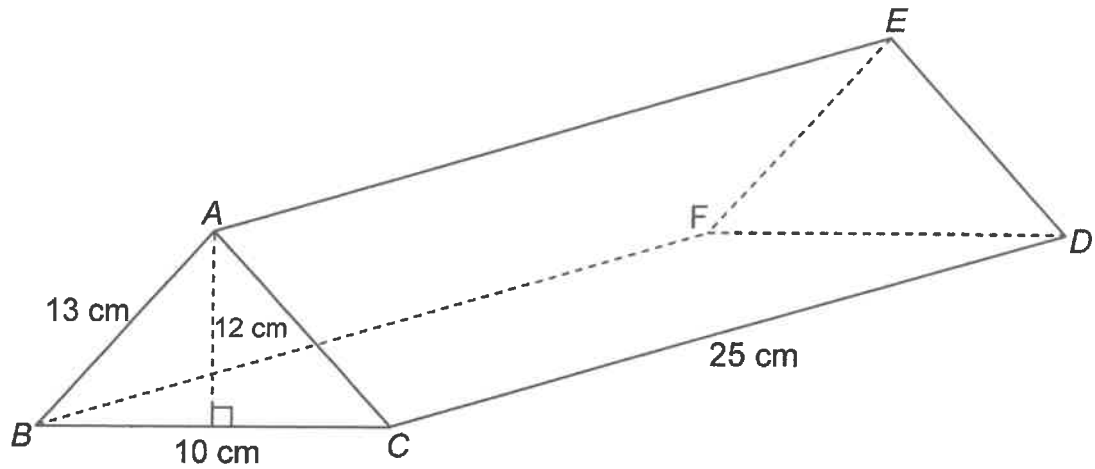
- (a) Use the grid below to draw the graph of the equation $y = x^2 - x$. (3)



- (b) On the grid above, draw the graph of $y = 1$. (1)
- (c) Use the graphs drawn in (a) and (b) to find the values of x that satisfy the equations $y = x^2 - x$ and $y = 1$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ (2)

21. The diagram below shows a triangular prism $ABCDEF$.



(a) Calculate the area of triangle ABC .

..... cm^2 (2)

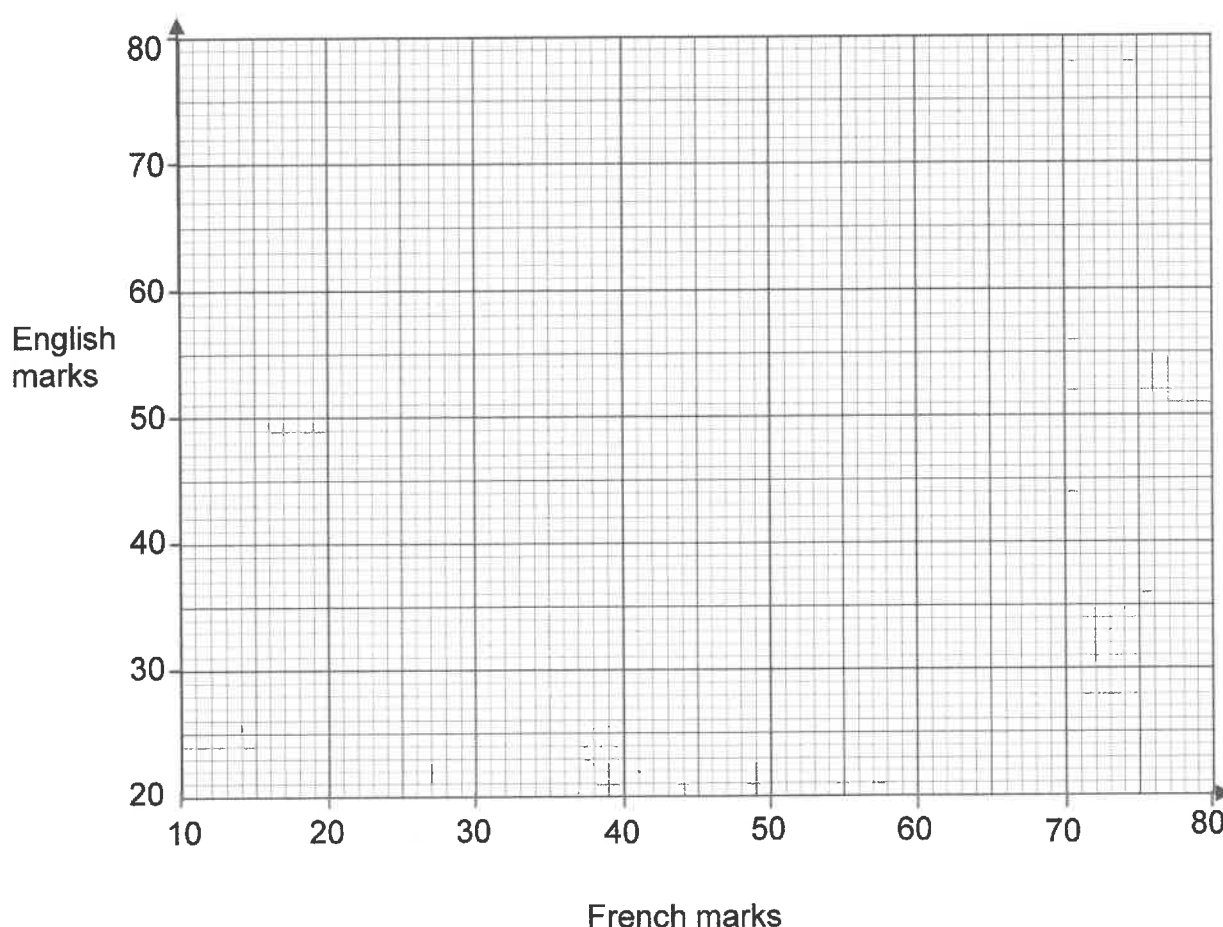
(b) Calculate the volume of the prism $ABCDEF$.

..... cm^3 (2)

22. Ten students sat for examinations in French and English. The table below shows the set of marks scored by each student on the examinations.

French	30	70	68	46	70	54	72	46	60	50
English	40	74	70	50	66	60	78	54	66	54

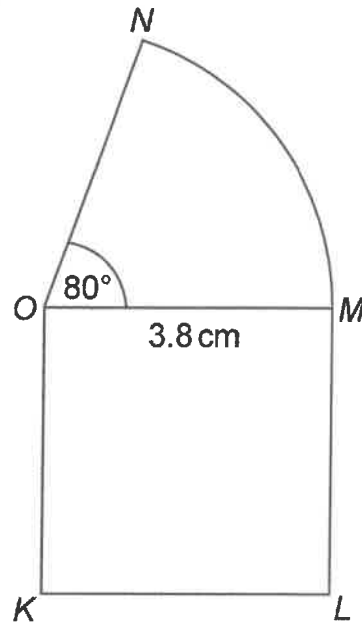
- (a) In the pair of axes below, plot the points for a scatter diagram of the marks scored in French and English by the ten students. (3)



- (b) Draw a line of best fit for the scatter diagram. (1)
- (c) A student who was absent during the English examination scored 65 marks in the French examination. Use your line of best fit to estimate the mark they would have scored in the English examination.

..... (1)

23. The diagram below shows a composite shape of a square $OKLM$ and a sector OMN of a circle, centre O .



- (a) Calculate the length of the arc MN .
Take π as 3.142 .

.....cm (2)

- (b) Calculate the perimeter of the composite shape $OKLMN$.

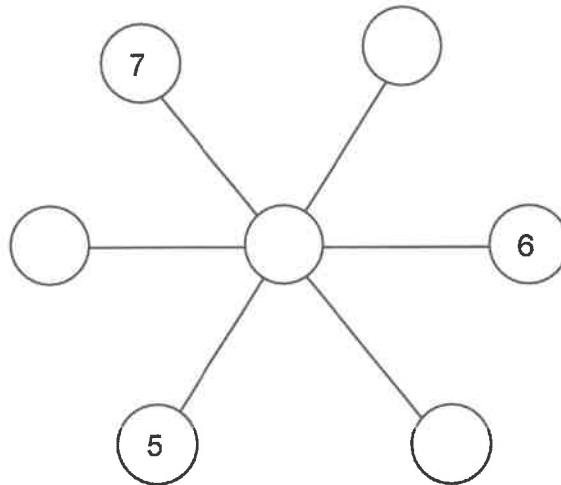
.....cm (2)

SECTION C

(20 Marks)

Answer **all** questions in this section.

24. The circles in the diagram below are to be filled with numbers 1, 2, 3, 4, 5, 6 and 7. Each number should appear only once. The sum of the numbers in the three circles on a straight line should be 10. The numbers 5, 6 and 7 have been filled for you. Fill in the missing numbers.



(3)

25. The shapes ,  and  represent different whole numbers. Given that;

$$\square + \bigcirc = 10$$

$$\triangle + \triangle = 6$$

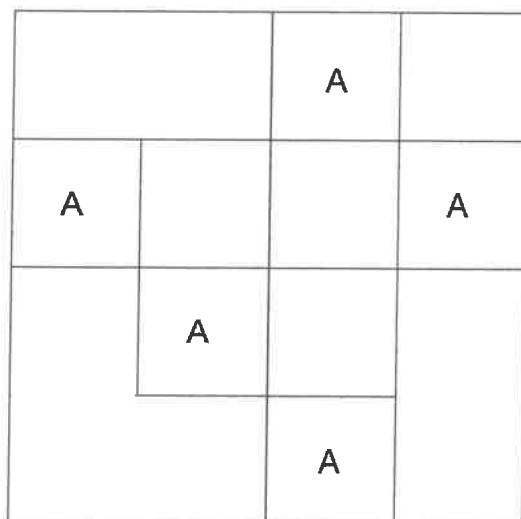
$$\triangle + \bigcirc = 5$$

What numbers are represented by the shapes?

$$\square = \dots\dots\dots \quad \triangle = \dots\dots\dots \quad \bigcirc = \dots\dots\dots$$

(3)

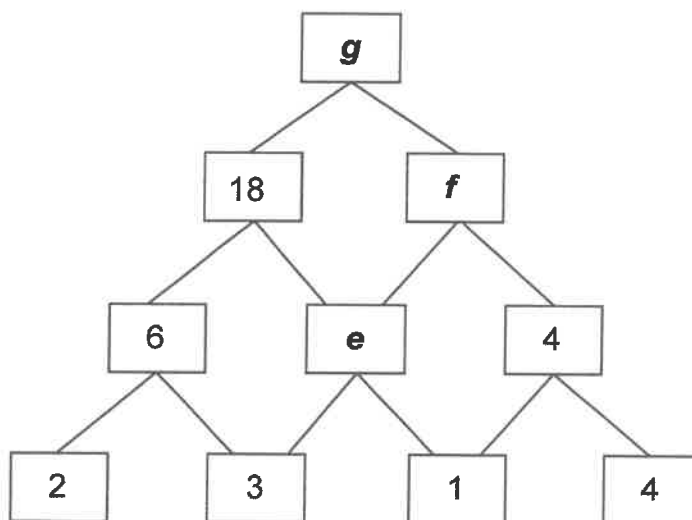
26. The diagram below is made of squares. Some of the identical squares are labelled A.



What is the total number of squares in the diagram?

..... (3)

27. Study the diagram below and write down the values of **e**, **f** and **g**.



e =

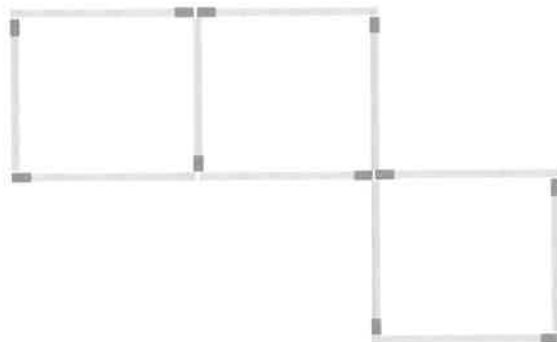
f =

g =

(3)

28. The diagram below shows how eleven matchsticks can be arranged to form three squares.

Three matchsticks in the diagram can be rearranged to form **two** squares of different sizes.



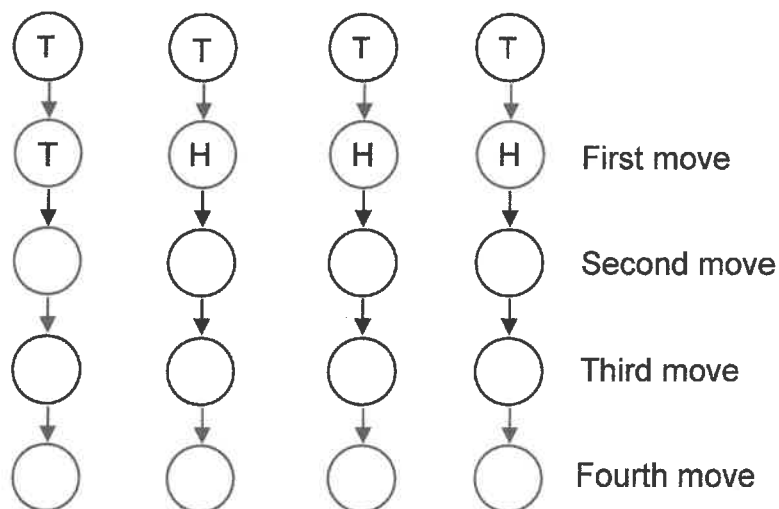
Draw the diagram of the **two** squares after the rearrangement of the three matchsticks.

(2)



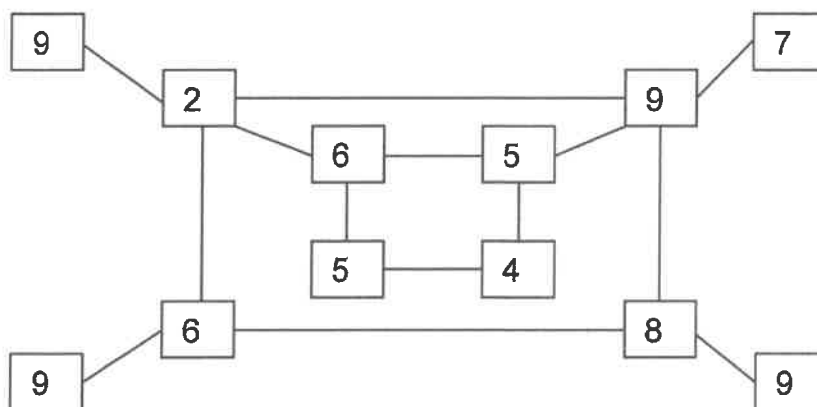
29. A game is played using four coins. Each coin has two sides, head (H) and tail (T). At the start of the game the coins are placed in a row, tails up. A move consists of turning three different coins once. The result of the first move is shown in the diagram below.

Complete the diagram for the second, third and fourth move such that the coins are all heads up in the fourth move.



(3)

30. The diagram below shows numbers connected by lines. Five numbers have to be chosen such that there is no skipping of a number or going back over a line. The five numbers chosen should give the maximum possible sum.

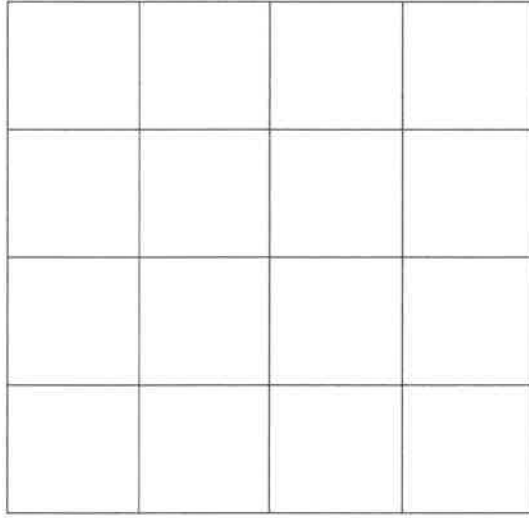


Write down the five numbers chosen and their sum.

=

(2)

31. Place 8 ticks (✓) inside the 4×4 square below such that each row, each column and each diagonal contains **only** two ticks. No square should contain more than one tick.



(1)



BOTSWANA EXAMINATIONS COUNCIL
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MATHEMATICS

13/2

Paper 2

October/November 2019

Marks: 100

Time: 2 Hours

**Candidate
Full Names:**

Centre Number:

J	C				
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Candidate Number:

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Section	Marks Scored
A	
B	
C	
Total Marks	

This question paper contains 16 printed pages.

Answer all questions in this section.

1. Calculate $\begin{pmatrix} 3 & 5 \\ -1 & 7 \end{pmatrix} + \begin{pmatrix} 11 & -4 \\ 8 & 5 \end{pmatrix}$.

 $\begin{pmatrix} & \\ & \end{pmatrix}$

(2)

2. Factorise $12mn - 4n$ completely.

(2)

3. Make t the subject of the formula $k = pt + 7$.

(2)

4. A piece of wood of length 6.5 m is cut into 40 equal pieces.

Calculate the length of each piece.

.....m (2)

5. Calculate the value of v for which $v = \frac{g+h}{5}$ when $g = 12$ and $h = 17$.

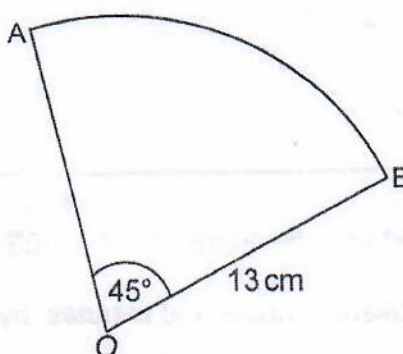
(1)

6. The total amount of money shared by Dineo and Monale was P750. Dineo's share was $\frac{3}{5}$ of the money.

How much, in Pula, was Monale's share?

P..... (2)

7. The diagram below shows a sector AOB of radius 13 cm and sector angle of 45° . Take π as 3.142.



Calculate the area of the sector AOB .

.....cm² (2)

8. Solve simultaneously the equations $6p - 4r = 14$ and $p - 2r = 5$.

$p = \dots\dots\dots$ $r = \dots\dots\dots$ (3)

9. The points A (7,4) and B (0,3) lie on a straight line.

What is the equation of the straight line?

..... (3)

10. Solve the equation $\frac{x-5}{3} = 7$.

..... (2)

11. A motorbike travelled at an average speed of 105 km/h for a distance of 245 km.

Calculate the time taken, in hours and minutes, by the motorbike to cover the distance of 245 km.

..... (2)

12. Express $\frac{3}{x+1} + \frac{4}{x}$ as a single fraction.

..... (3)

13. Solve the equation $x^2 - 13x + 30 = 0$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ (3)

14. Factorise $u^2 - 121$ completely.

$\dots\dots\dots$ (1)

282

A003



SECTION B

(50 Marks)

Answer **all** questions in this section.

15. The time in Johannesburg, South Africa, is 10 hours ahead of the time in California, USA. An aircraft leaves Johannesburg on Monday at 1750hrs and flies to California.

(a) Express 1750 in 12-hour clock notation.

..... (1)

(b) What was the time in California when the aircraft left Johannesburg?

..... (2)

The time in California was 0850 on Tuesday when the aircraft arrived.

(c) How much time was the journey from Johannesburg to California?

..... (2)

282

A003



16. A triangle XYZ is such that $XY = 8.5$ cm and angle $XZY = 58^\circ$.

(a) Use the line YZ below to construct triangle XYZ.



(3)

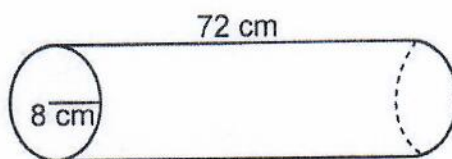
(b) Measure and write down the actual length of the side XZ.

.....cm (1)

(c) Construct a perpendicular bisector of the line YZ.
(Show all lines of construction).

(2)

17. The diagram below represents a wooden prism. The prism has a length of 72 cm and radius 8 cm.



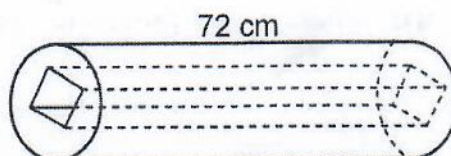
- (a) What is the name of the prism?

..... (1)

- (b) Calculate the volume of the prism.

.....cm³ (2)

A hole, in the shape of a cuboid, is drilled out from the wooden prism as shown in the diagram below. The cuboid has a cross-sectional area of 15 cm².



- (c) Calculate the volume of the wood that has been drilled out to make the hole.

.....cm³ (2)

- (d) Calculate the volume of the **remaining** wood.

.....cm³ (2)

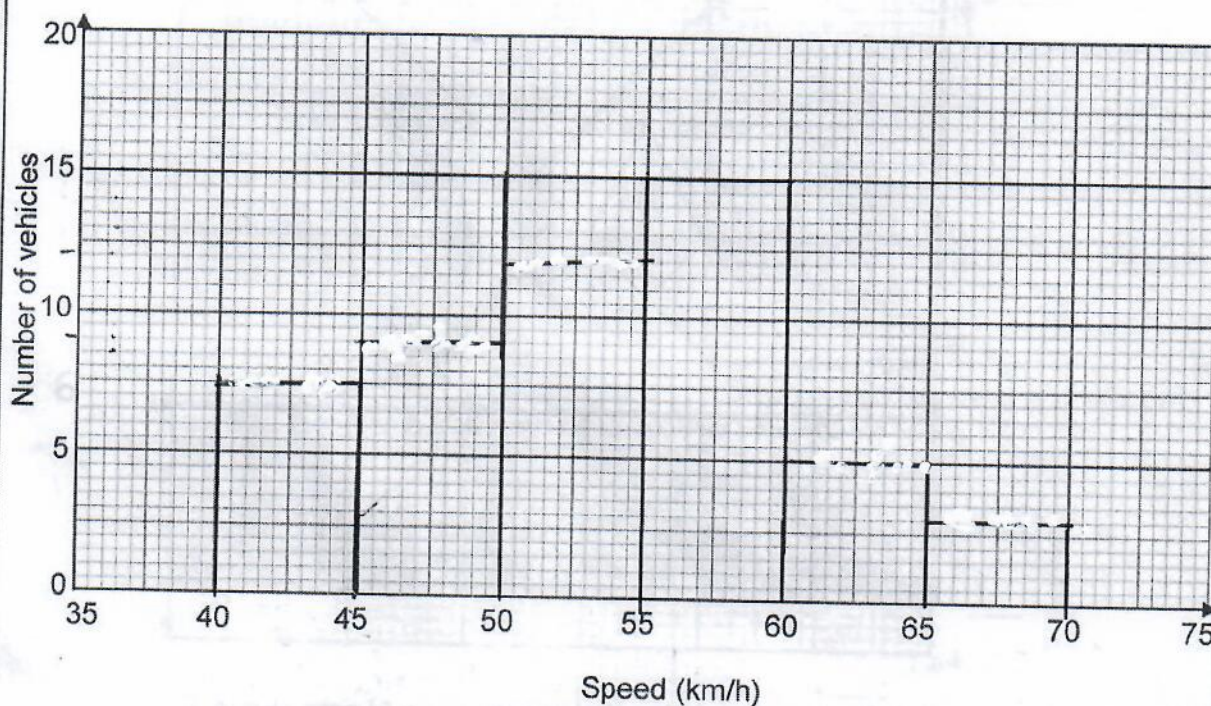
18. The speed, in km/h, of vehicles passing in front of a school gate was recorded. The results are shown in the table below.

Speed (km/h)	40-45	45-50	50-55	55-60	60-65	65-70
Number of vehicles	6	8	12	15	5	3

- (a) What is the modal class?

..... (1)

- (b) On the grid below, draw a histogram to represent the speed of the vehicles.

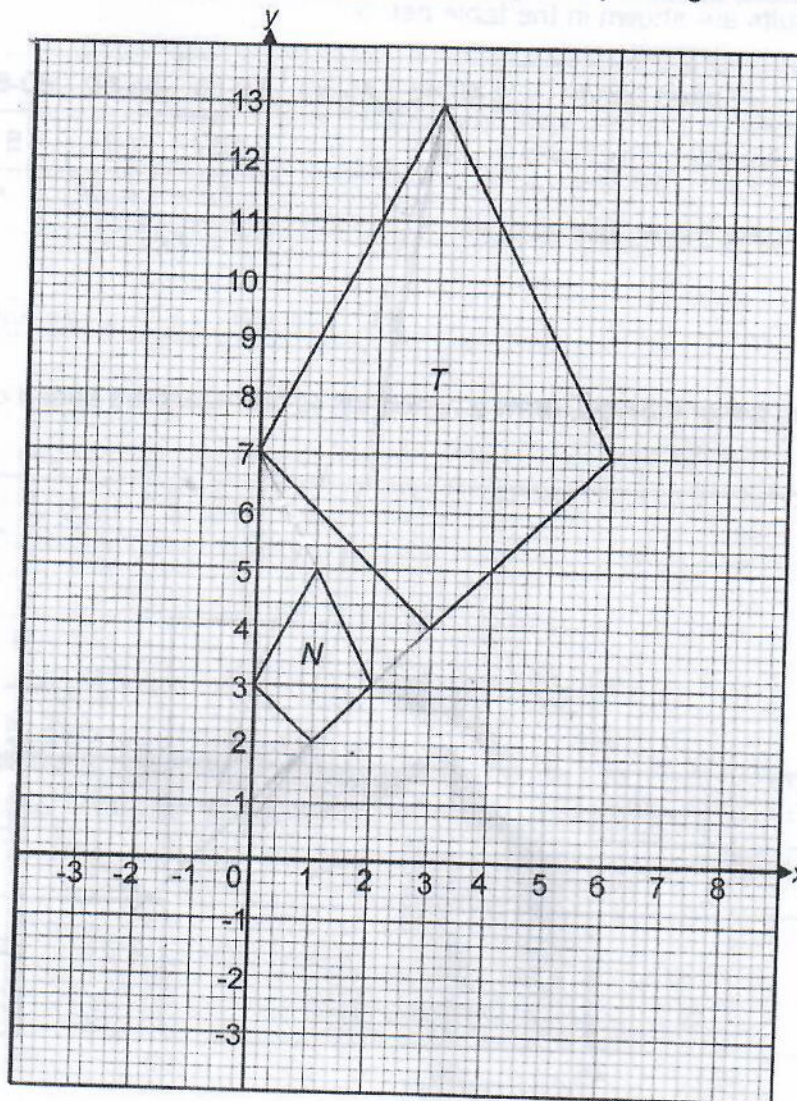


- (c) On the same grid above, draw a frequency polygon for the speed of the vehicles.

- (d) The speed limit of the vehicles passing in front of the school gate was 55 km/h. How many vehicles were over speeding?

..... (1)

19. In the diagram below, shape N is mapped onto shape T by a single transformation.

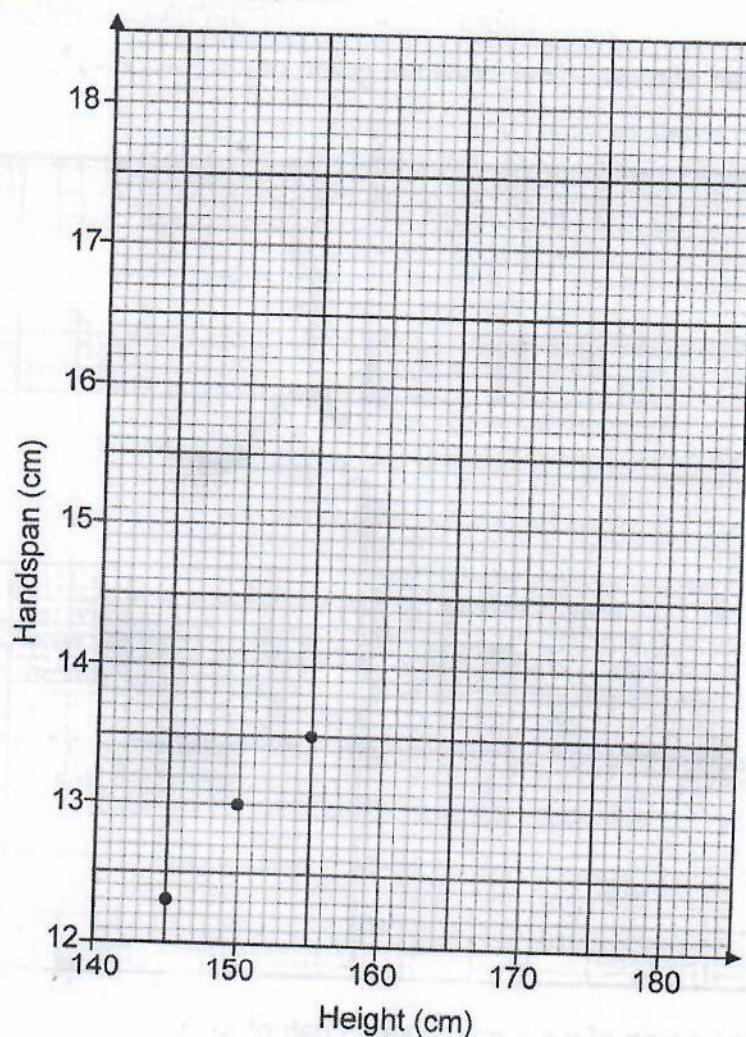


- (a) Describe fully the transformation that maps shape N onto shape T .
.....
..... (3)
- (b) On the grid above, draw and label shape D with vertices $(1, 0)$, $(2, -1)$, $(1, -3)$ and $(0, -1)$. (2)
- (c) Describe fully the transformation that maps shape N onto shape D .
.....
..... (2)

20. A volleyball team has 10 players. The team coach measured each team member's height and the handspan, that is, the maximum distance between the tips of the thumb and the little finger. The results are shown in the table below.

Height (cm)	145	150	155	152	162	160	160	168	170	180
Handspan (cm)	12.3	13	13.5	14	14	14.5	15	15.3	16.5	16.8

The two measures are to be shown in a scatter diagram. The first three points have been plotted for the scatter diagram.



- (a) Complete the scatter diagram by plotting the remaining 7 points. (2)
- (b) Draw a line of best fit on the scatter diagram. (1)
- (c) Using your line of best fit, estimate the handspan of a player with a height of 169 cm.

.....cm (1)

21. The table below shows values of x and corresponding values of y for the equation $y = 3 + 2x - x^2$.

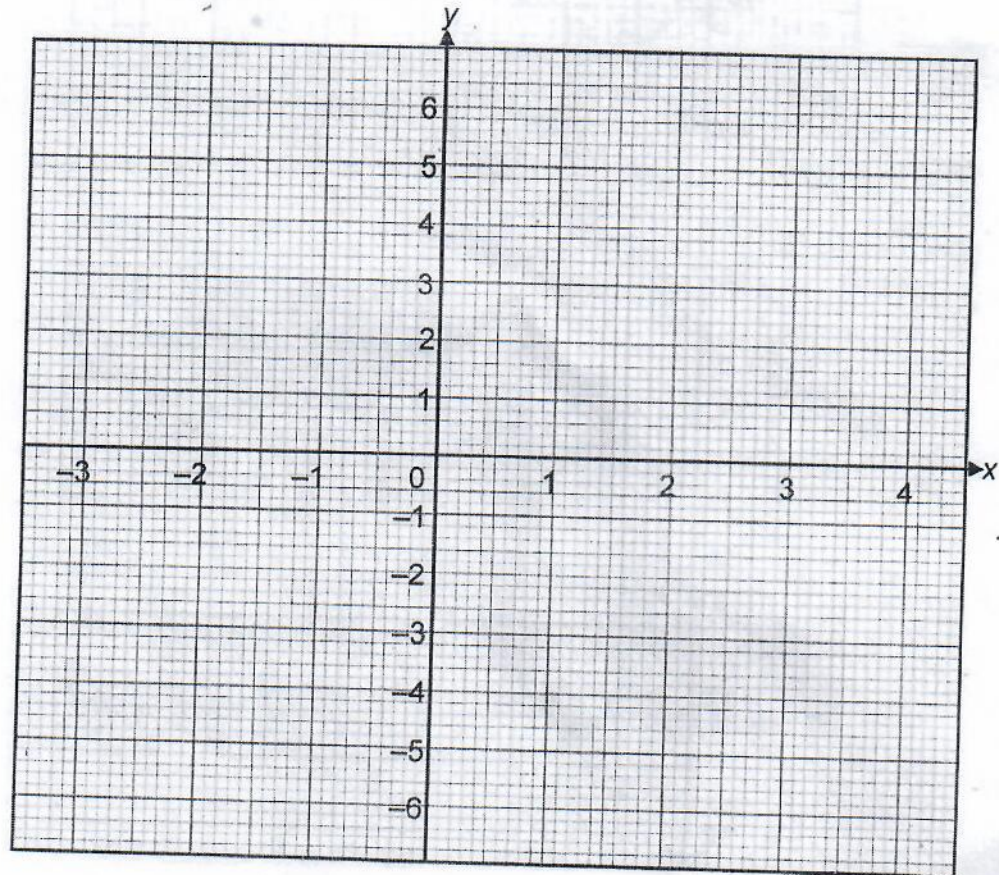
x	-2	-1	0	1	2	3	4
y	-5	0	3	4	3	0	p

- (a) What is the value of p ?

..... (1)

- (b) On the pair of axes below, draw the graph of $y = 3 + 2x - x^2$.

(3)



- (c) Draw the graph of $y = 2$ on the same pair of axes.

(1)

- (d) Use the graph to find the values of x that satisfy the equation $3 + 2x - x^2 = 2$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ (2)

22. The table below shows the different interest rates charged for loans given by a motshelo scheme.

Loan Amount in Pula (P)	Interest rate per month (%)
100 – 7500	8
7500 – 15000	12
15000 – 20000	14

- (a) What is the interest rate for a loan of P8000?

.....% (1)

- (b) Lebogang is given a loan of P17 000.

- (i) Calculate the amount of money that Lebogang will be charged as interest at the end of one month?

P..... (2)

The scheme requires a customer to pay interest and 25% of the loan at the end of each month.

- (ii) What is the **total** amount of money that Lebogang will be required to pay at end of the first month?

P..... (3)

SECTION C

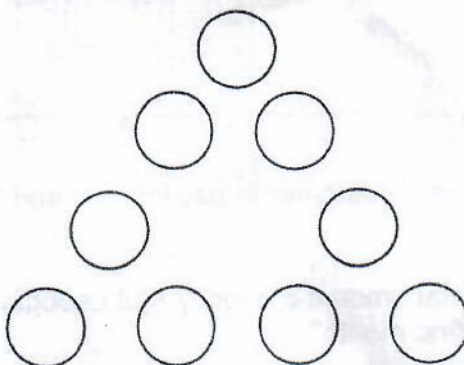
(20 Marks)

Answer **all** the questions in this section.

23. How many multiples of 3 are there which are less than 100 but **not** divisible by either 9 or 10?

..... (3)

24. Write the numbers 1 up to 9 in the circles that form a triangle as shown below. The sum of numbers on each side of the triangle should be 20. Each number must be used once.



..... (3)

25. Ndipo's age is 3 years plus half her age.

What is Ndipo's age?

..... (2)



26. There are two trucks A and B. The trucks are loaded with bags of maize. If one bag of maize is taken from truck A and loaded onto truck B, then they will have the same number of bags. If a bag was taken from truck B and loaded to truck A then the bags in truck A will be twice the number of bags in truck B.

How many bags of maize are in truck A and in truck B?

Truck A =

Truck B = (3)

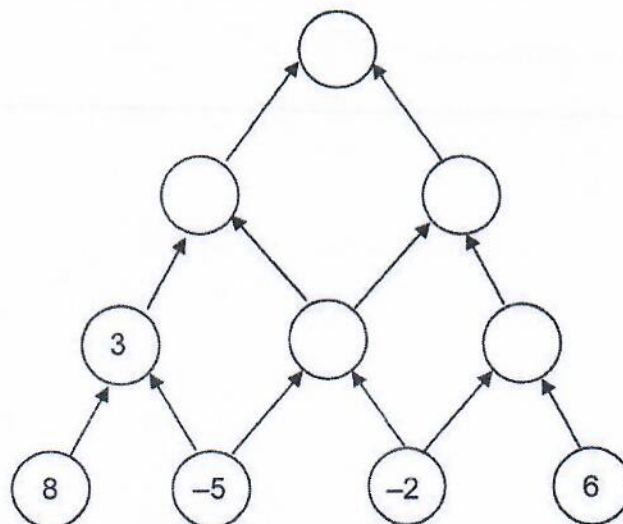
27. A teacher has more than two types of pens in her bag. All of the pens are blue, except for two pens. All of them are red, except for two pens. They are all black except for two pens.

How many of each type of pen does the teacher have in her bag?

Blue.....Black.....Red..... (3)

28. In the number pyramid below, each number in the upper level is obtained by adding the numbers in the lower level.

Fill in the missing numbers to complete the number pyramid.



(3)



MATHEMATICS 4

Term 1-Revision Questions



JANUARY 1, 2019

.....BY CT MON.....

1. $135 + 30 =$
 - A. 234
 - B. 165
 - C. 135
 - D. 531

2. $203 - 37 =$
 - A. 203
 - B. 167
 - C. 166
 - D. 991

3. Write 48 in words.
 - A. Fouty eighty
 - B. Fouty eight
 - C. Forty eight
 - D. Eighy four

4. What is the place value of 3 in 2392?
 - A. Hundred
 - B. Tens
 - C. Unit
 - D. Thousand
 - E.

5. Mpho has 60 bananas. He shared them among his 6 friends .how many bananas will each one get?
 - A. 3
 - B. 10
 - C. 14
 - D. 5

6. $135+601+111=$

- A. 874
- B. 847
- C. 999
- D. 1000

USE THE NUMBERLINE BELOW TO ANSWER

7. Which number is nearer 20?

- A. 10
- B. 16
- C. 18
- D. 81

8. Which number is near to 104?

- A. 100
- B. 99
- C. 110
- D. 99

9. Which number is missing in 100 200 ----- 400 500

- A. 400
- B. 199
- C. 300
- D. 299

10. Which of the following numbers are multiples of 4?

- A. 16,14,24
- B. 3,6,9
- C. 7,9,11
- D. 4,8,9

11. Which of the following are multiples of 3?

- A. 3,9,11,12,
- B. 21,24,27,30
- C. 9,12,15,19,21
- D. 3,6,7,11

12. What is the missing number

8, 10, 12, ----- 16

- A. 16
- B. 18
- C. 14
- D. 11

13. Write the place value of the underline numbers in 967

- A. Tenths
- B. Units
- C. Hundreds
- D. Thousands

USE THE SYMBOLS = < >

14.

- A. 85 ----- 59
- B. 117 ----- 171
- C. 999-----1000
- D. 119 ----- 191
- E. 987----- 789

15. WRITE THE MISSING NUMBERS IN THE SEQUENCE BELOW

- A. 10, 20, 30, 40, -----, 60
- B. 400 500 -----, 700,800
- C. 15, 20, 25, ----, 30

16. Work out the following

- A. $73 + 123 =$
- B. $145 - 10$
- C. $32 \times 3 =$ -----
- D.
$$\begin{array}{r} + 547 \\ 433 \\ \hline \end{array}$$

- E. $16 + 20 + 122 =$ -----

SOLVE THE PROBLEM BELOW

- 17. There are 525 children in Lorwana Primary and 205 in Thamaga . **How** many children are there **altogether** ----- **(show your working) 2**
- 18. Kago had 30 sweets and she share them among 10 children.
How many sweets did each child get? (show your working) 2
- 19. Richmark had 300 chickens and sold 20 to Spar supermarket. How many chickens are they left with -----? **(show your work) (2)**

20. There are 410 pupils in a school, 198 are boys .**How many are girls in this school? (show your working) 2**

21.. Mma Seboko had planted 46 mango trees and 22 peaches. **What is the total number of trees grown? ----- (show your working (2)**

22. In a class four there are 8 children. Each child has 15. **How many pencils are there altogether? (show your working) 2 marks**

23. USE LIST OF NUMBERS BELOW TO ANSWER QUESTION

(1, 2, 3, 4, 5, 6, 7, 8, 9)

- A. Which numbers are multiples of 3-----?
- B. Which numbers are even -----
- C. Which ones are factors of 12-----
- D. Which numbers are even numbers?-----

24. What is the place value of the underlined digit?

- A. 8193
- B. 9465

25. Work out the following sums

- A. $33 \times 10 = \text{-----}$
- B. $5 \times 10 = \text{-----}$
- C. $135 \times 3 = \text{-----}$
- D. $123 \times 4 = \text{-----}$
- E. $44 \times 5 = \text{-----}$

26. Work out the following sums.

- A. $\text{----} \times 4 = 20$
- B. $2 \times 12 = \text{-----}$
- C. $100 \div \text{-----} = 20$
- D. $500 \div 10 = \text{-----}$
- E. $400 \div 10 = \text{-----}$
- F. $907 \div 9 = \text{-----}$

27. Work out

A.
$$\begin{array}{r} 100 \\ + 323 \\ \hline 347 \\ \hline \text{-----} \end{array}$$

B.
$$\begin{array}{r} 1000 \\ - 909 \\ \hline \text{-----} \end{array}$$

C.
$$\begin{array}{r} 123 \\ + 150 \\ \hline 67 \\ \hline \text{-----} \end{array}$$

D.
$$\begin{array}{r} 142 \\ \times 3 \\ \hline \end{array}$$

28. Find factors of the following numbers (2marks)

- A. 24: -----
- B. 32: -----

29. Complete the following pattern(3marks)

66, 63, 60, ----- 54, 51, -----, -----

30. WORK OUT

- A. $75 + 30 =$ -----
- B. $135 + 500 =$ -----
- C. $900 - 248 =$ -----
- D. $1000 - 519 =$ -----

31. WORK OUT

- A. $24 \times 10 =$ -----
- B. $385 \times 10 =$ -----
- C. $752 \times 10 =$ -----

32. WORK OUT

A.
$$\begin{array}{r} 205 \\ 9 \overline{) } \end{array}$$

B.
$$\begin{array}{r} 35 \\ 7 \overline{) } \end{array}$$

30. Which of the numbers are even?

17, 12, 22, 41, 19

31. Write two hundred and twenty one as a numeral -----
32. Write one thousand as a numeral -----
33. 705 in words is -----
34. One thousand six hundred and thirty two in numerals is -----
35. 1085 in words is-----

80 revision Questions

.....Good luck STD 4`s.....



BOTSWANA EXAMINATIONS COUNCIL
JUNIOR CERTIFICATE EXAMINATION

MATHEMATICS

Paper 2

13/2

October/November 2018

Marks: 100

Time: 2 Hours

Candidate
Full Names:

Centre Number:

J	C				
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Candidate Number:

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INSTRUCTIONS

1. Write your examination number in the space provided above.
2. Answer **ALL** questions.
3. **ALL** answers must be written in the spaces provided.
4. Show all the necessary working.
5. Calculators may be used in this paper.
6. Use the value π in your calculator unless otherwise stated.
7. If the degree of accuracy is **not** specified in the question and the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

FOR EXAMINER'S USE ONLY

Section	Marks Scored
A	
B	
C	
Total Marks	

SECTION A

(30 Marks)

Answer all questions in this section.

1. Write down the two prime numbers between 50 and 60.

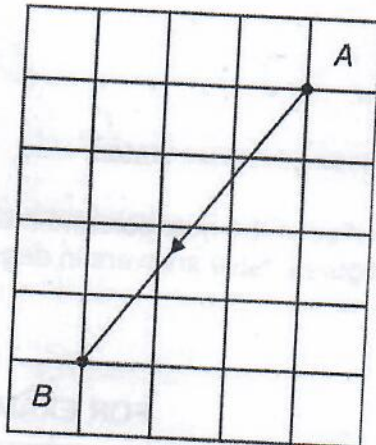
..... (2)

2. A box contains sweets of the same size, shape and texture. However, there are 3 different flavours. There are 2 chocolate, 3 mint and 7 strawberry flavoured sweets in the box.

Calculate the probability of randomly picking a strawberry flavoured sweet from the box.

..... (2)

3. Vector AB is shown on the grid below.



Write down $\vec{AB}$ as a column vector.

$$\vec{AB} = \begin{pmatrix} \\ \end{pmatrix} \quad (1)$$

4. Solve $5(x - 4) = 3x - 7$.

$x = \dots\dots\dots$ (3)

5. The value of a car depreciated from P90 000 in June 2015 to P76 500 in June 2016.
Calculate the percentage depreciation of the value of the car between June 2015
and June 2016.

$\dots\dots\dots$ (2)

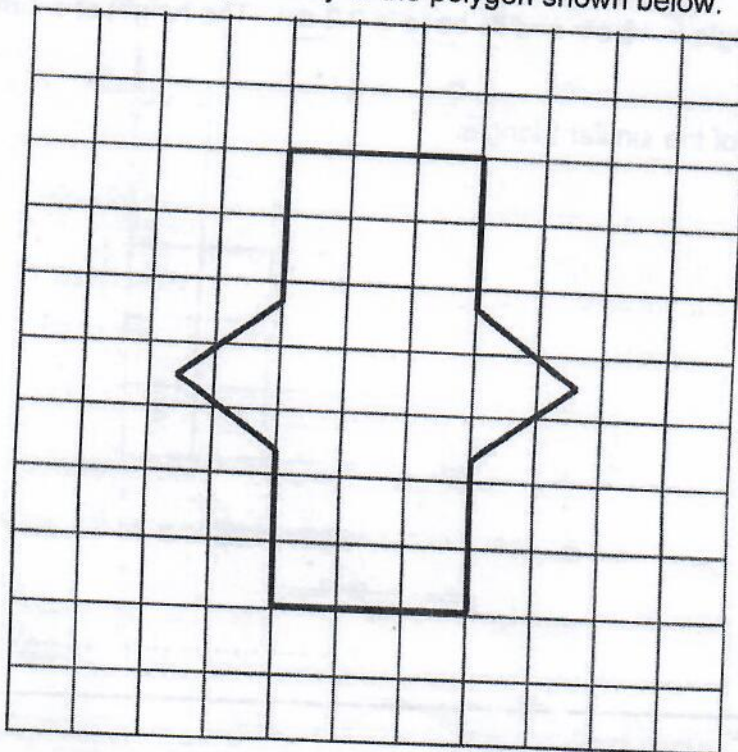
6. The height of a triangle is 16 cm and its base is 9.2 cm. The height of a similar
triangle is 12 cm.
Calculate the base of the similar triangle.

$\dots\dots\dots$ (2)

7. A 4 m ladder leans against a vertical wall at an angle of 50° from a horizontal ground. One end of the ladder rests on the top edge of the vertical wall.
- What is the height of the vertical wall?

..... (2)

8. Draw all the lines of symmetry of the polygon shown below.



(2)

9. A circle has an area of 59.45 cm^2 .

Calculate the radius of the circle.

Take π as 3.142.

..... (3)

10. Solve the following simultaneous equations.

$$x + 2y = 9$$

$$y = 3x + 1$$

$x = \dots\dots\dots, y = \dots\dots\dots$ (3)

11. A sequence of numbers is given as;

-3, 1, 5, 9, 13, ...

- (a) Write down the next two terms of the sequence.

..... (2)

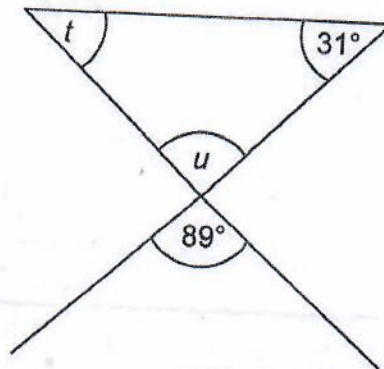
- (b) What is the common difference for the sequence?

..... (1)

- (c) Write down the general formula, in terms of n , for the sequence.

..... (1)

In the diagram below, the size of two angles is given and the size of the other two angles is marked as t and u . Use the diagram to answer question 12.



12. (a) State, with a reason, the size of the angle marked u .

.....

..... (2)

- (b) Calculate the size of the angle marked t .

..... (2)

SECTION B

(50 Marks)

Answer **all** questions in this section.

13. A car started a journey at 11:20 am and completed it at 2:35 pm.

(a) Write down 2:35 pm using 24 hour clock notation.

..... (1)

(b) Calculate the time, in hours and minutes, that the car took to complete the journey.

.....hours.....minutes. (2)

(c) The total distance covered during the journey was 234 km.

Calculate the average speed, in km/h, of the car.

..... km/h (2)

14. A farmer had 250 chickens to sell at a market. The cost of one chicken was P48.

(a) He sold 8% of the chickens in the morning.

Calculate the number of chickens that the farmer sold in the morning.

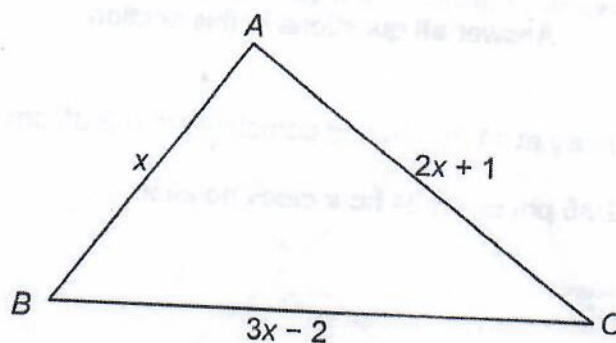
..... (2)

(b) The farmer then sold 36 chickens in the afternoon.

Calculate the total amount of money made from the sale of the chickens in the morning and in the afternoon.

..... (2)

15. Triangle ABC is shown below.



- (a) Form an expression, in terms of x , for the perimeter of triangle ABC .

..... (1)

- (b) The perimeter of the triangle is 35 cm.

Form an equation, in terms of x , for the perimeter of the triangle.

..... (1)

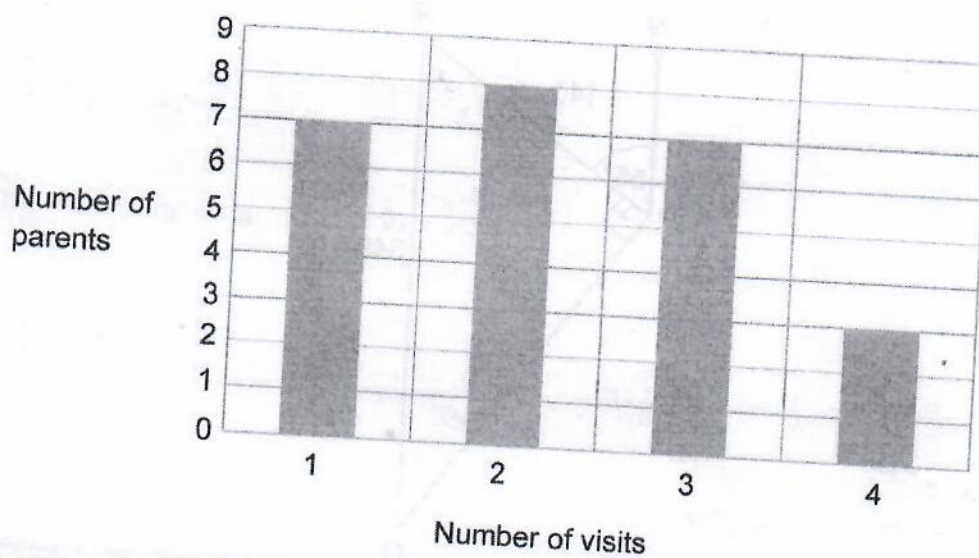
- (c) Solve the equation in 15 (b).

..... (2)

- (d) Calculate the length of side AC .

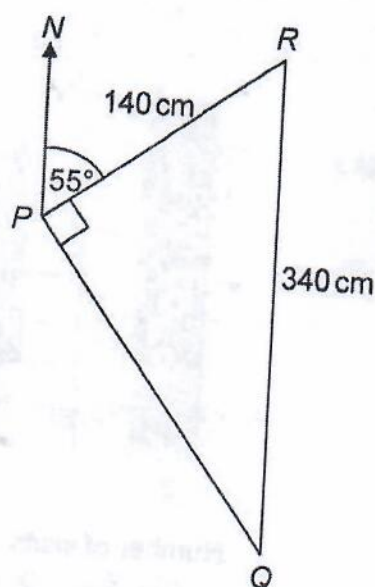
..... (1)

16. The bar chart below shows the number of visits made by 25 parents to a school.



- (a) State the modal number of visits.
..... (1)
- (b) Calculate the total number of visits by the parents.
..... (2)
- (c) Calculate the mean number of visits by the parents.
..... (2)

17. The diagram below shows the positions of three points P , Q and R .



- (a) What is the bearing of point Q from point P ?
..... (1)
- (b) Calculate the size of angle PRQ .
..... (2)
- (c) Calculate the distance between point P and point Q .
..... (2)

18. (a) Factorise $x^2 + 9x + 20$ completely.

..... (2)

- (b) Solve the equation $x^2 + 9x + 20 = 0$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ (2)

19. Labour charge, in Pula, for repairing any machine for h hours is given as $L = 300 + 120h$. The labour charge for repairing a photocopying machine was P1056.

How many hours were spent on repairing the machine?

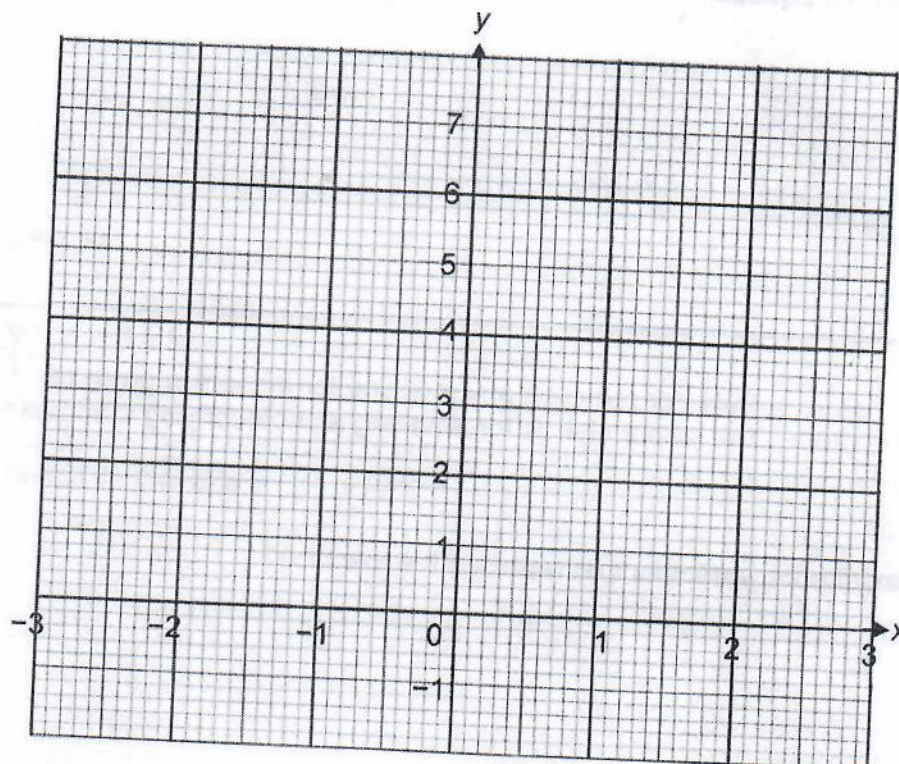
..... (3)

20. The table below shows values of x and corresponding values of y for the equation $y = x^2 - x$.

x	-2	-1	0	1	2	3
y	6	2	0	0	2	6

- (a) Use the grid below to draw the graph of the equation $y = x^2 - x$.

(3)



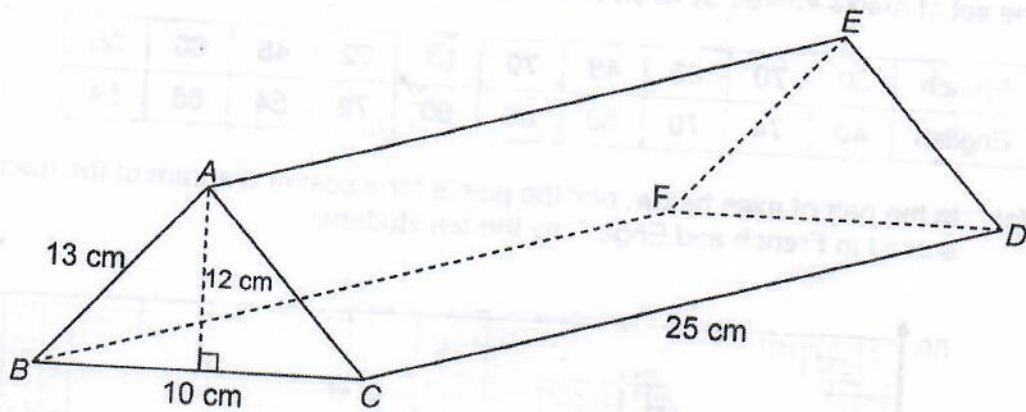
- (b) On the grid above, draw the graph of $y = 1$.

(1)

- (c) Use the graphs drawn in (a) and (b) to find the values of x that satisfy the equations $y = x^2 - x$ and $y = 1$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ (2)

21. The diagram below shows a triangular prism $ABCDEF$.



- (a) Calculate the area of triangle ABC .

..... cm^2 (2)

- (b) Calculate the volume of the prism $ABCDEF$.

..... cm^3 (2)

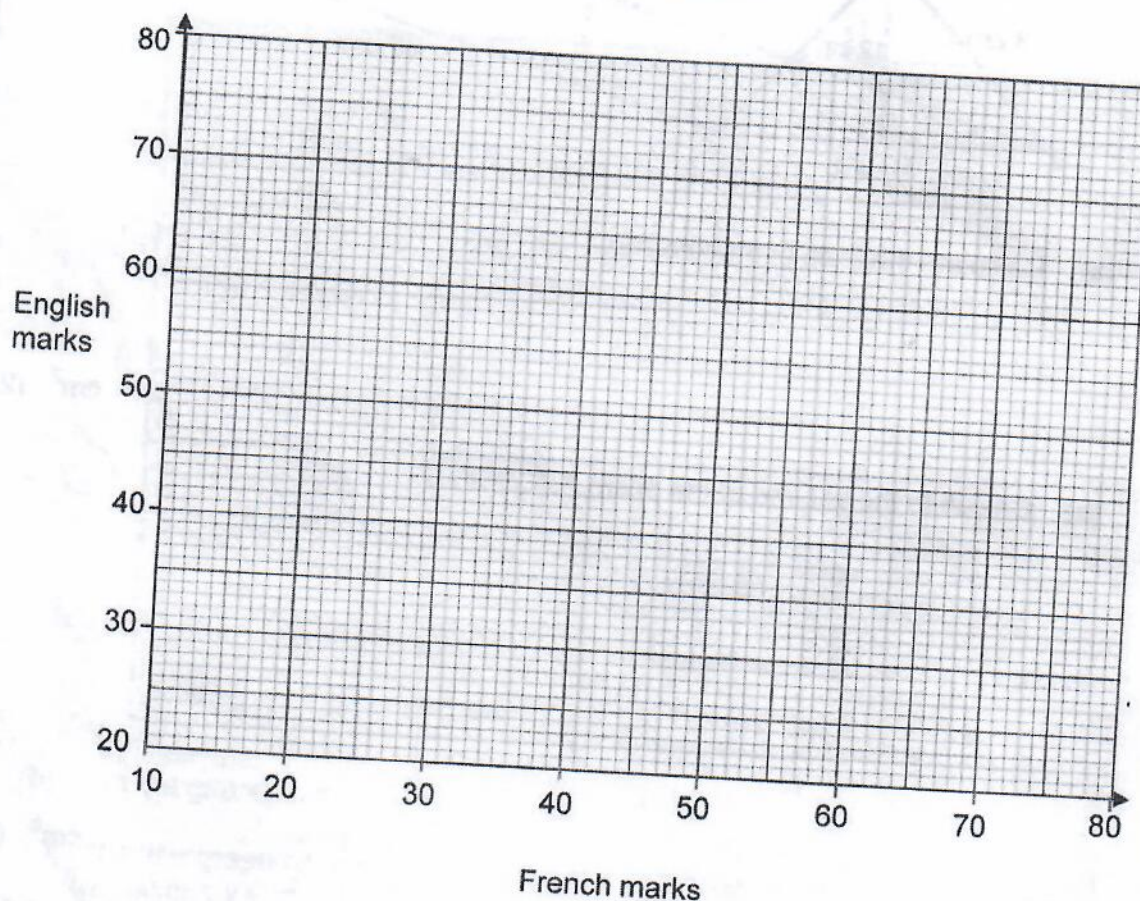


22. Ten students sat for examinations in French and English. The table below shows the set of marks scored by each student on the examinations.

French	30	70	68	46	70	54	72	46	60	50
English	40	74	70	50	66	60	78	54	66	54

- (a) In the pair of axes below, plot the points for a scatter diagram of the marks scored in French and English by the ten students.

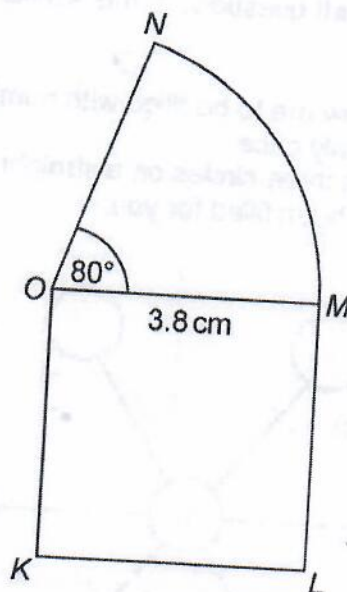
(3)



- (b) Draw a line of best fit for the scatter diagram. (1)
- (c) A student who was absent during the English examination scored 65 marks in the French examination. Use your line of best fit to estimate the mark they would have scored in the English examination.

..... (1)

23. The diagram below shows a composite shape of a square $OKLM$ and a sector OMN of a circle, centre O .



- (a) Calculate the length of the arc MN .
Take π as 3.142.

.....cm (2)

- (b) Calculate the perimeter of the composite shape $OKLMN$.

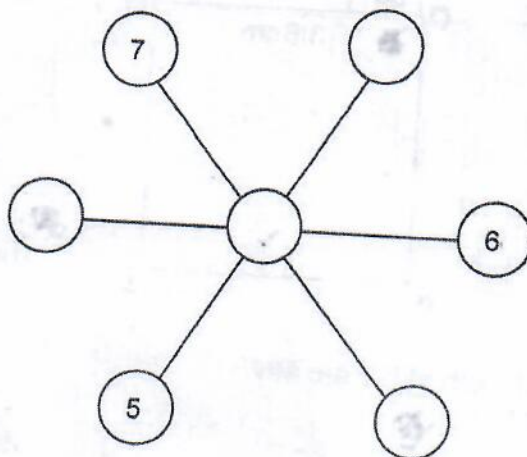
.....cm (2)

SECTION C

(20 Marks)

Answer all questions in this section.

24. The circles in the diagram below are to be filled with numbers 1, 2, 3, 4, 5, 6 and 7. Each number should appear only once. The sum of the numbers in the three circles on a straight line should be 10. The numbers 5, 6 and 7 have been filled for you. Fill in the missing numbers.



(3)

25. The shapes ,  and  represent different whole numbers. Given that;

$$\square + \bigcirc = 10$$

$$\triangle + \triangle = 6$$

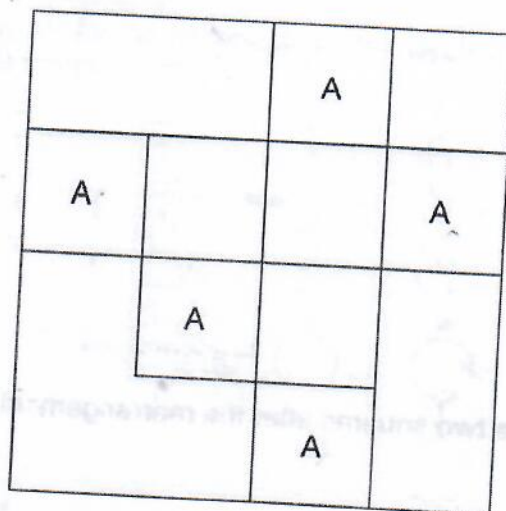
$$\triangle + \bigcirc = 5$$

What numbers are represented by the shapes?

$$\square = \dots\dots\dots \triangle = \dots\dots\dots \bigcirc = \dots\dots\dots$$

(3)

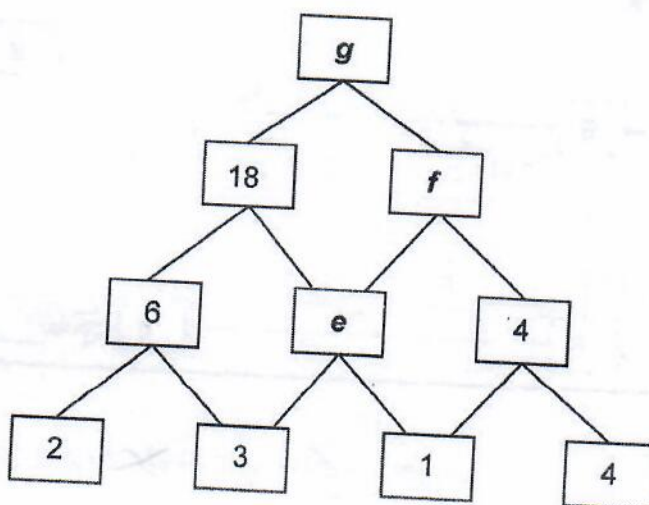
26. The diagram below is made of squares. Some of the identical squares are labelled A.



What is the total number of squares in the diagram?

..... (3)

27. Study the diagram below and write down the values of e , f and g .



$e =$

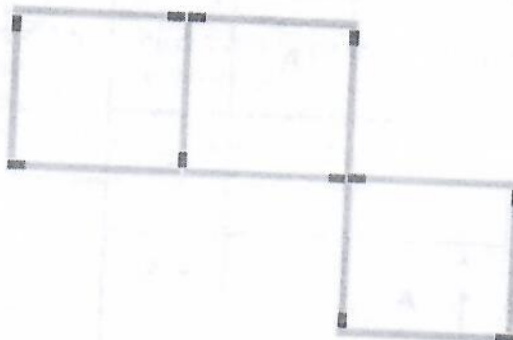
$f =$

$g =$

(3)

28. The diagram below shows how eleven matchsticks can be arranged to form three squares.

Three matchsticks in the diagram can be rearranged to form **two** squares of different sizes.



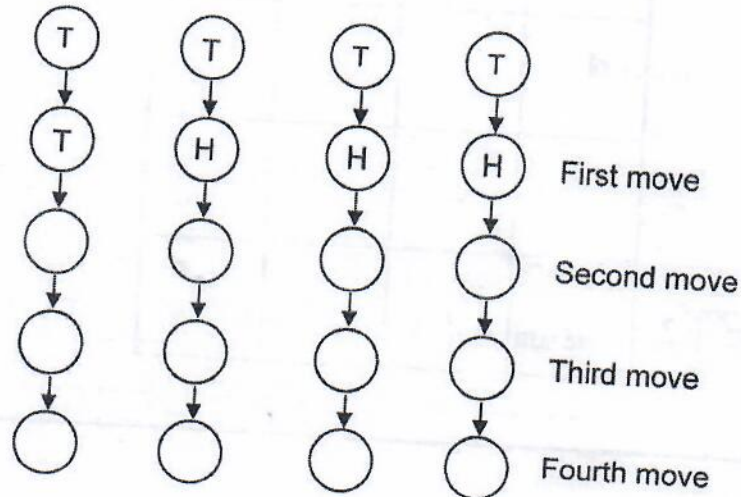
Draw the diagram of the **two** squares after the rearrangement of the three matchsticks.



(2)

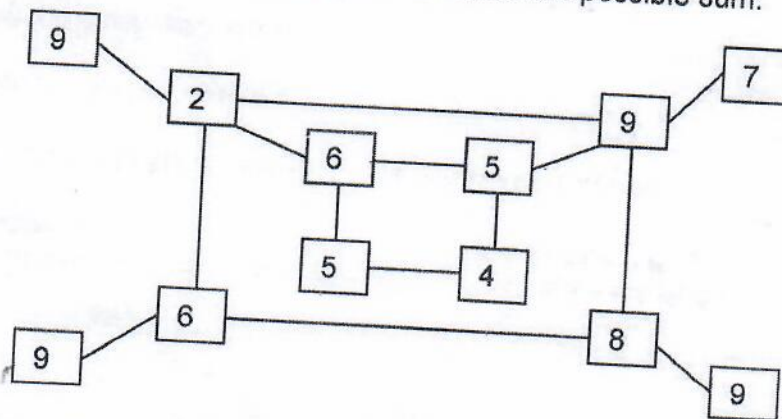
29. A game is played using four coins. Each coin has two sides, head (H) and tail (T). At the start of the game the coins are placed in a row, tails up. A move consists of turning three different coins once. The result of the first move is shown in the diagram below.

Complete the diagram for the second, third and fourth move such that the coins are all heads up in the fourth move.



(3)

30. The diagram below shows numbers connected by lines. Five numbers have to be chosen such that there is no skipping of a number or going back over a line. The five numbers chosen should give the maximum possible sum.



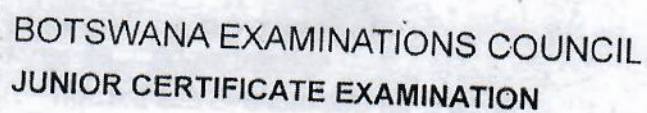
Write down the five numbers chosen and their sum.

=

(2)

31. Place 8 ticks (✓) inside the 4×4 square below such that each row, each column and each diagonal contains **only** two ticks. No square should contain more than one tick.

(1)



13/2

Paper 2

October/November 2017

Marks: 100

Time: 2 Hours

**Candidate
Full Names:**

--

Centre Number:

J	C				
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Candidate Number:

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INSTRUCTIONS

1. Write your full names and examination number in the spaces provided above.
2. Answer **ALL** questions.
3. **ALL** answers must be written in the spaces provided.
4. Show all the necessary working.
5. Calculators may be used in this paper.
6. Use the value π in your calculator unless otherwise stated.
7. If the degree of accuracy is **not** specified in the question and the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

FOR EXAMINER'S USE ONLY

Section	Marks Scored
A	
B	
C	
Total Marks	

This question paper contains 18 printed pages.

DO NOT TURN THE PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

(30 Marks)

For
Examiner's
Use

Answer all questions in this section.

1. (a) Evaluate $25 - 7 \times 4 + 6$.

..... (1)

- (b) Express 0.74 as a percentage.

.....% (1)

2. A biscuit manufacturing company packs 26 boxes per carton. How many of these boxes can be packed in 2 cartons?

..... (2)

3. Simplify $7(x - 2) + 3x$.

..... (2)

4. Simplify $\frac{20m^2p^7}{5mp^3}$.

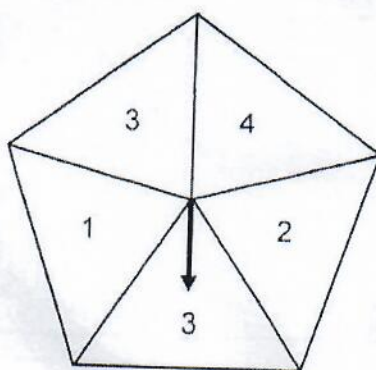
..... (2)



5. Express $\frac{x}{4} + \frac{3x}{7}$ as a single fraction.

..... (2)

6. The diagram below shows a spinner with 5 equal triangles numbered 1, 3, 4, 2, 3 and an arrow. When the arrow is spun, it finally stops in one of the five triangles.



Calculate the probability that the arrow will stop at a triangle either numbered 3 or 4.

..... (2)

7. Express T in terms of A in the formula $A = \frac{3}{2+T}$.

..... (3)

8. The distribution below shows the mass of 6 rugby players in kg.

102, 109, 124, 126, 135, 141

- (a) What is the range of the mass distribution?

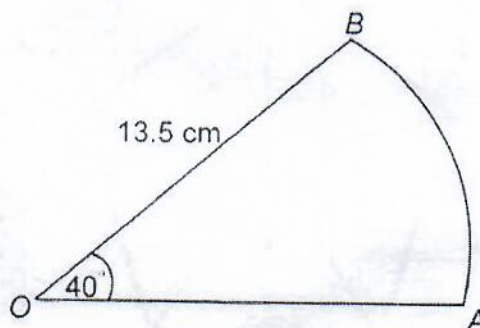
.....kg (1)

- (b) Explain why the distribution has no modal mass?

.....

..... (1)

9. The diagram below shows a sector of radius 13.5 cm and angle BOA of 40° .

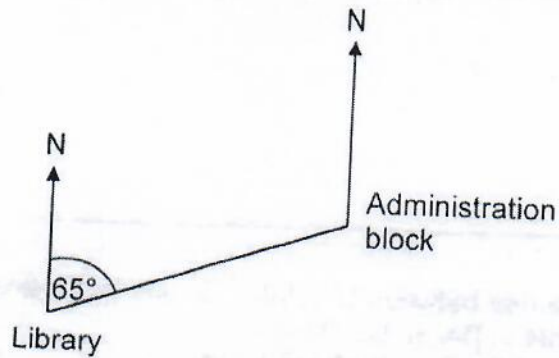


Calculate the length of the arc AB.

.....cm (2)



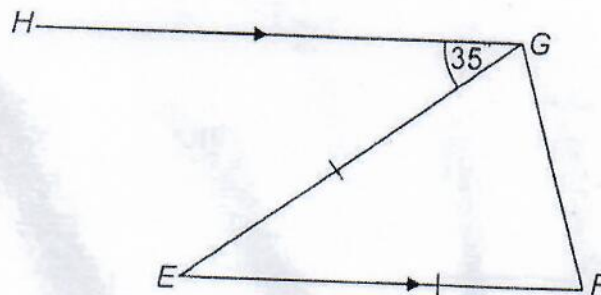
10. The diagram below shows the positions of the library and the administration block in a school. The bearing of the administration block from the library is 065° .



Calculate the bearing of the library from the administration block.

..... (3)

11. The diagram below shows triangle EFG and line HG parallel to EF . The line EG is equal to side EF and angle EGH is 35° .



- (a) State with a reason the size of angle GEF .

..... (2)

- (b) Calculate the size of angle EFG .

..... (2)

12. A bus started a journey at 5:32 pm and arrived after 1 hour 41 minutes.

At what time did the bus arrive?

..... (2)

13. The exchange rate between US Dollar (\$) and Botswana Pula (P) was
P1 = \$0.112994 in December 2015.
How much was P2000 worth in US Dollars (\$)?

..... (2)



SECTION B

(50 Marks)

Answer **all** questions in this section.

14. The table below shows the times, in seconds, taken to download different games using a mobile phone.

Name of game	Time (seconds)
Karate Kid	35
Football Kickoff	78
Brick Destroyer	120
Tennis Champion	48
Bolt Challenge	210

The cost of downloading each game is 6 thebe per second.

- (a) Express the time, in minutes, taken to download Bolt Challenge.

..... (1)

- (b) What is the total cost of downloading a Football Kickoff game and a Karate Kid game one after the other?

..... (2)

- (c) Which game will cost **exactly** P7.20 to download?

..... (2)

15. The grouped frequency table below shows the length of time in a day, in minutes, taken by 200 people to watch television.

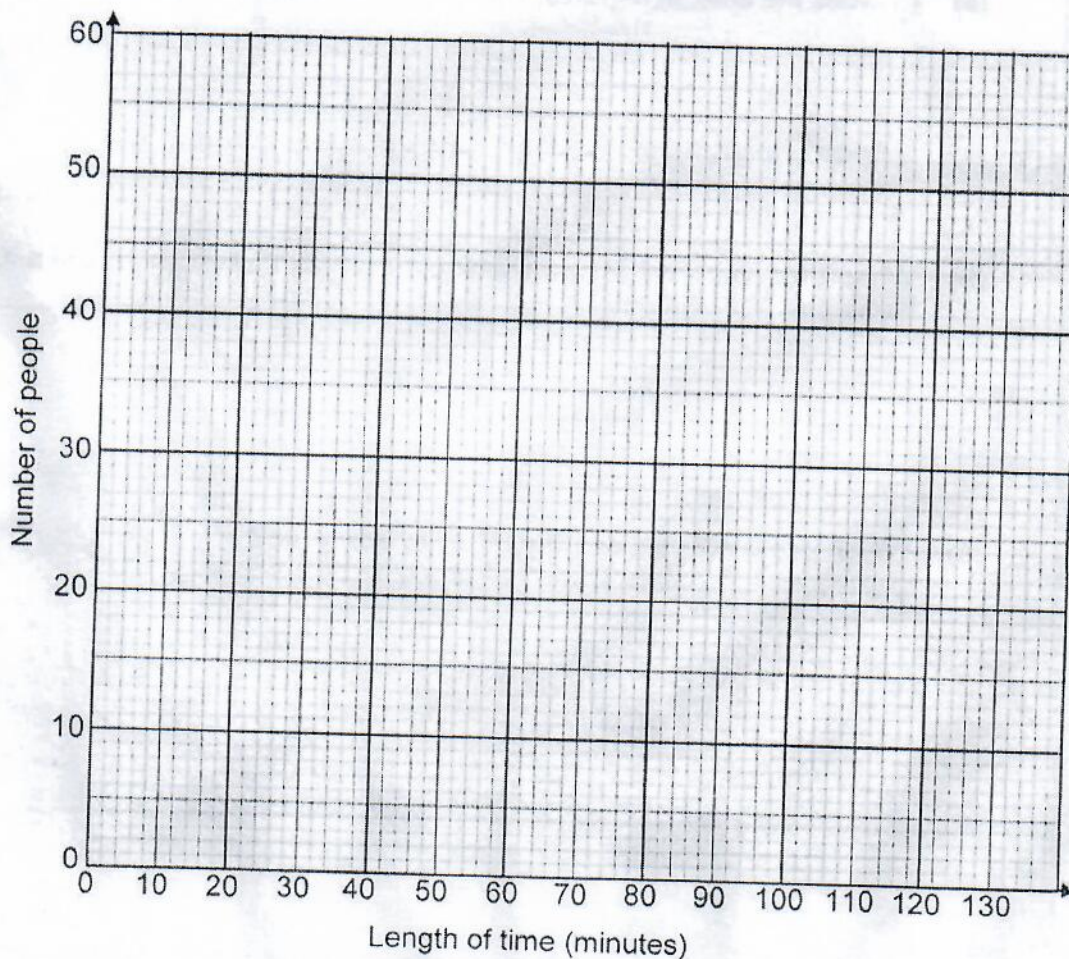
Length of time (minutes)	Number of people
$0 \leq x < 20$	10
$20 \leq x < 40$	32
$40 \leq x < 60$	48
$60 \leq x < 80$	54
$80 \leq x < 100$	36
$100 \leq x < 120$	20

- (a) What is the median class?

(1)

- (b) On the grid below, draw a histogram to represent the information given in the table.

(3)



16. A stock brick costs Px and a face brick costs Py at Kago Bricks Company. Mr Molefe buys 200 stock bricks.

(a) (i) Express, in terms of x , the cost of the 200 stock bricks.

..... (1)

(ii) He also buys 300 face bricks.

Express, in terms of y , the cost of the 300 face bricks.

..... (1)

(b) The total cost of the stock bricks and face bricks bought by Mr Molefe is P1050. Write down an equation, in terms of x and y , for the total cost of the bricks.

..... (1)

(c) Mrs Tag also bought 50 stock bricks and 100 face bricks for a total cost of P325 from Kago Bricks Company.

(i) Form an equation, in terms of x and y , for the total cost of the bricks bought by Mrs Tag.

..... (2)

(ii) Show that the equation in (c) (i) reduces to $2x + 4y = 13$.

..... (1)

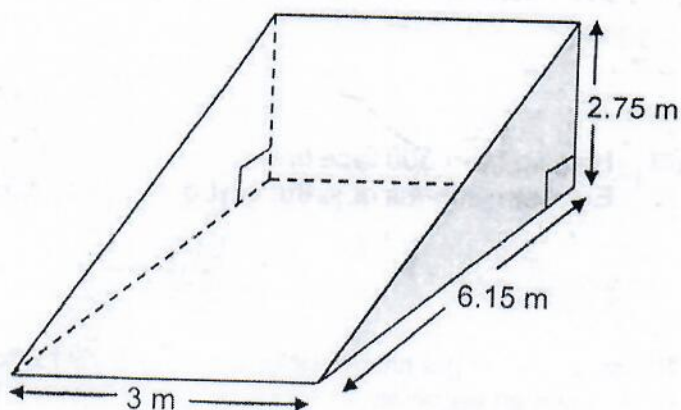
(d) Solve the equations in 16 (b) and $2x + 4y = 13$ simultaneously.

..... (3)

(e) How much does one stock brick at Kago Bricks Company cost?

..... (1)

17. The diagram below represents a motor cycle ramp in the form of a prism with a shaded face.



- (a) What is the name of the prism?

..... (1)

- (b) Calculate the area of the shaded face.

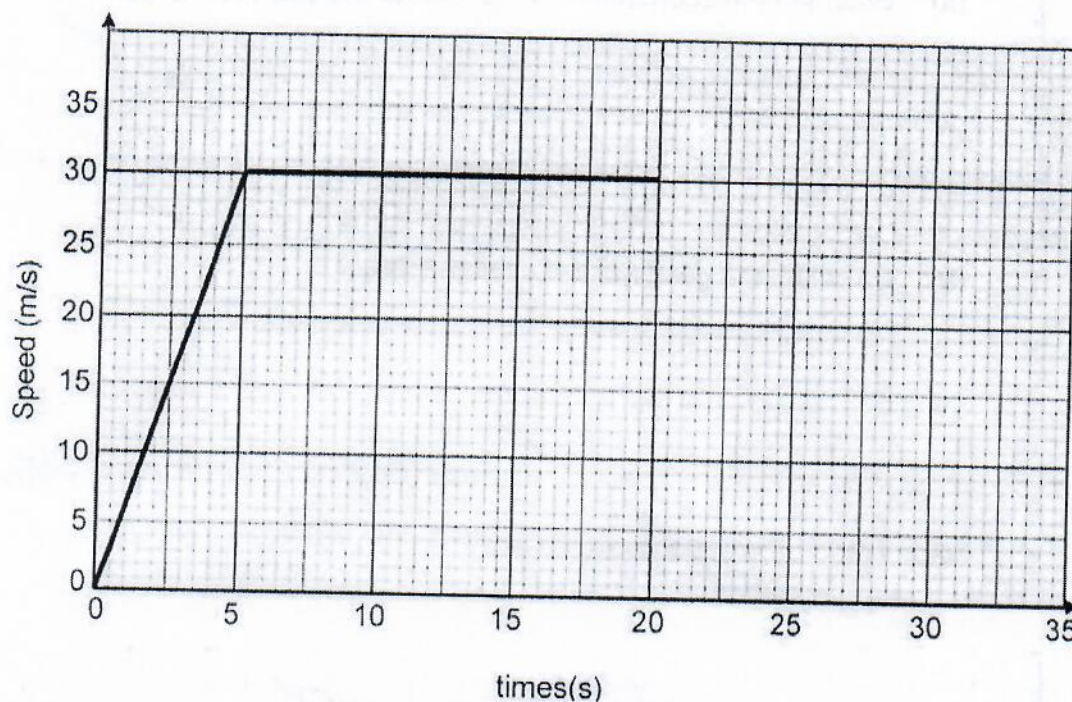
..... (2)

- (c) An identical ramp is to be constructed using concrete mixture.
Calculate the volume of the concrete mixture needed to construct the ramp.

..... (2)



18. A cyclist had a training session. The results obtained for the first 20 seconds of the training are represented in an incomplete speed-time graph shown below.



- (a) Calculate the acceleration during the first 5 seconds.

..... (2)

- (b) Calculate the distance travelled during the first 20 seconds.

..... (3)

- (c) After 20 seconds, the cyclist decreased his speed constantly from 30 m/s to 0 m/s in 7.5 seconds.

Use this information to complete the graph above.

(1)

19. A straight line passes through the points $K(3,11)$ and $M(1,7)$.

(a) What are the coordinates of the midpoint of point K and point M ?

(.....,) (2)

(b) Calculate the gradient of the straight line.

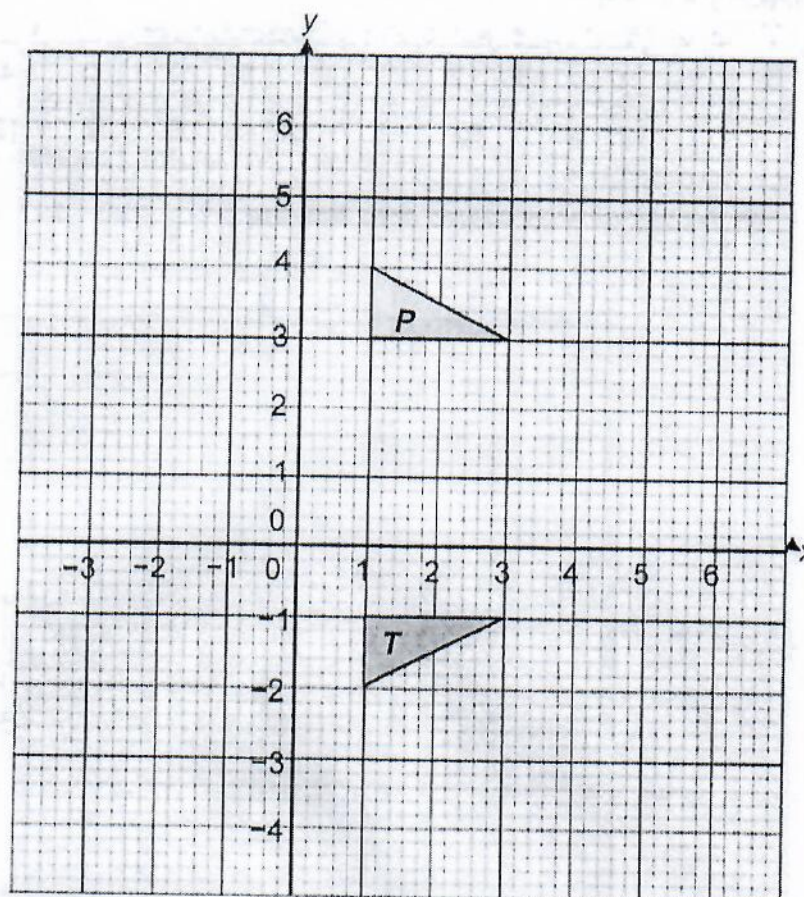
..... (2)

(c) What is the equation of the line?

..... (2)



20. In the diagram below, triangle T is mapped onto triangle P after a transformation.



- (a) Describe fully the transformation that maps triangle T onto triangle P .

.....
..... (2)

- (b) Triangle P is mapped onto triangle Q by a translation through vector $\begin{pmatrix} -3 \\ -2 \end{pmatrix}$.
On the grid above, draw and label triangle Q .

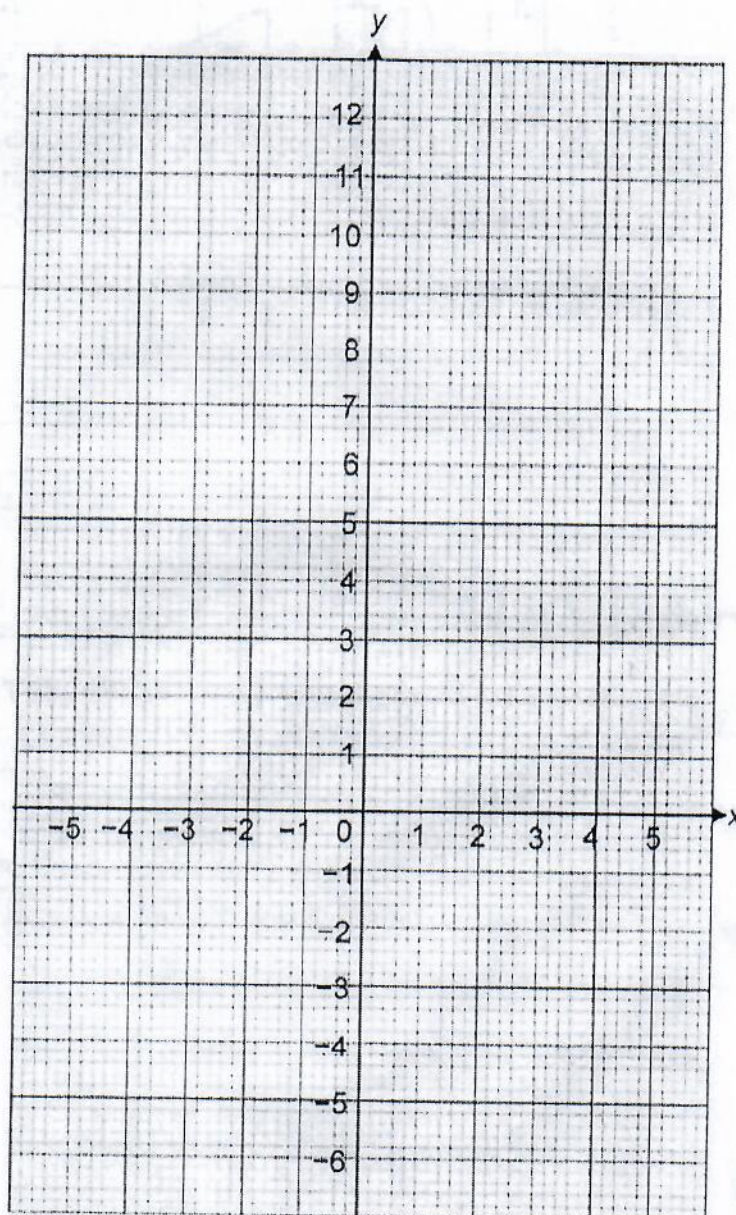
(2)

21. The table below shows some values of x and the corresponding values of y for the equation $y = x^2 - 4$.

x	-3	-2	-1	0	1	2	3	4
y	5	0	-3	-4	-3	0	5	12

- (a) On the pair of axes below, draw the graph of $y = x^2 - 4$.

(3)



- (b) On the same pair of axes, draw the line $y = 5$.

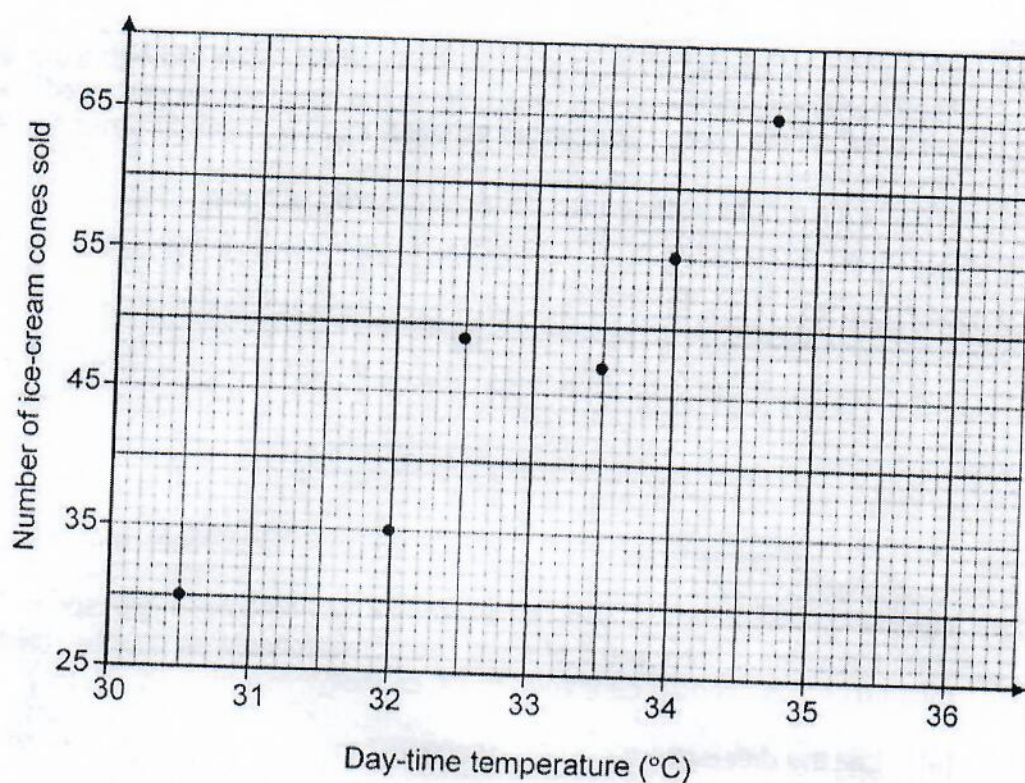
(1)

- (c) Use the graph to find the values of x that satisfies the equation $x^2 - 4 = 5$.

$x = \dots\dots\dots$ and $x = \dots\dots\dots$ (2)



22. A school tuck shop recorded the day-time temperatures and the number of ice-cream cones sold each day. The scatter diagram below shows the number of ice-cream cones sold against temperatures recorded.



- (a) Complete the table below to show the day-time temperatures recorded and the number of ice-cream cones sold.

(2)

Day-time temperature(°C)	Number of ice-cream cones sold
30.5	30
32	35
32.5	49

- (b) Draw a line of best fit on the scatter diagram above.

(1)

- (c) What type of correlation is shown by the scatter diagram?

..... (1)

SECTION C

(20 Marks)

Answer **all** the questions in this section.

23. Mr Sebitso measured a rectangular garden using a metal rod and a piece of string. The shorter side of the garden was 6 times the length of the metal rod but 4 times the length of the string. The longer side of the garden was 15 times the length of the metal rod.
What is the length of the garden when measured with the string?

..... (3)

24. Minibuses seating 10, 12 and 15 passengers can be used to transport 120 church choir members. There are 5 of each size of minibuses that may be used. All the minibuses used should be loaded to full capacity.

(a) List the different ways the passengers can be distributed in the minibuses.

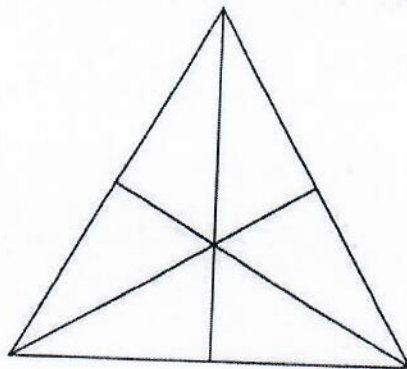
..... (2)

(b) What is the **least** number of minibuses that can be used?

..... (1)



25. How many triangles are there in the following diagram?



..... (3)

26. In a 24 hour period from midday to midnight, how many times are the minute hand and the hour hand of a clock at right angles to each other?

..... (2)

27. The volume of a certain mixture in a test-tube doubles each minute. In one hour, the mixture completely filled the test-tube.

After how many minutes was the test-tube half full?

..... (1)

28. What is the next number in the sequence below?

1, 2, 6, 42, 1806,

..... (2)

29. A woman has six children, and there are 4 years between each child and the next. She had her first child when she was 23 years old. Now the youngest of the six children is 19 years old.

How old is the woman?

..... (3)

30. A 3×3 square below has numbers 7, 13 and 37. Fill in the other squares with numbers such that numbers along the same row, same column and same diagonal each add up to 111.

		7
13	37	

..... (3)





BOTSWANA EXAMINATIONS COUNCIL
JUNIOR CERTIFICATE EXAMINATION

MATHEMATICS

13/2

Paper 2

October/November 2013

Marks: 100

Time: 2 Hours

Candidate's Examination Number:

Centre					Candidate				

INSTRUCTIONS

1. Write your examination number in the space provided above.
2. Answer **ALL** questions.
3. **ALL** answers must be written in the spaces provided.
4. Show all the necessary working.
5. Calculators may be used in this paper.
6. Use the value π in your calculator unless otherwise stated.
7. If the degree of accuracy is **not** specified in the question and the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

FOR EXAMINER'S USE ONLY

Section	Marks Scored
A	
B	
C	
Total Marks	

This question paper contains 16 printed pages.

SECTION A

(30 Marks)

Answer all questions in this section.

1. (a) How many sides does a pentagon have?

..... (1)

- (b) Calculate the sum of the interior angles of a pentagon.

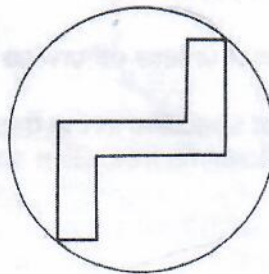
..... (2)

2. The time in Brazil is 5 hours behind the time in Botswana.
A live football game started at 17 00 in Brazil.

What time was it in Botswana when the game started?

..... (1)

3. What is the order of rotational symmetry of the diagram shown below?



..... (1)

4. Solve the equation $2y + 5 = 21$.

 $y =$ (2)

5. Express 125 km/h in m/s.

.....m/s (2)

6. Factorise $14xy - 21x + 7$ completely.

..... (2)

7. Solve the equation $x^2 - 3x - 15 = 0$.

$x =$ or (3)

Use the information below to answer question 8.

The number of people who voted in an election in Monwapo village in 2009 was 2 500. In 2012, the number of people who voted was 14% less than in 2009.

8. Calculate the number of people who voted in 2012.

..... (2)

9. It took 5 workers 6 hours to clean cars on Monday.
On Tuesday the number of workers was increased to 12.
How many hours will it take them to clean the same number of cars if they work at the same rate?

..... hours (2)

10. Express $\frac{3}{x+2} - \frac{1}{2}$ as a single fraction.

..... (3)

11. Make a the subject of the formula $v = u + \frac{1}{2}at$.

..... (3)

12. A garage services 27 buses and 18 trucks in one week.

Express the number of buses to the number of trucks as a ratio, in its simplest form.

..... (2)

13. The cost of one cow is P5 200 and the cost of one bull is P6 400. A farmer bought 43 cows and 2 bulls.

Calculate the total cost of cows and bulls that the farmer bought.

P..... (2)

14. In the formula $E = \frac{1}{2}mv^2$, calculate the value of E when $m = 19$ and $v = 13$.

..... (2)

Answer all questions in this section.

15. The first four terms of a sequence are 2, 5, 8 and 11.

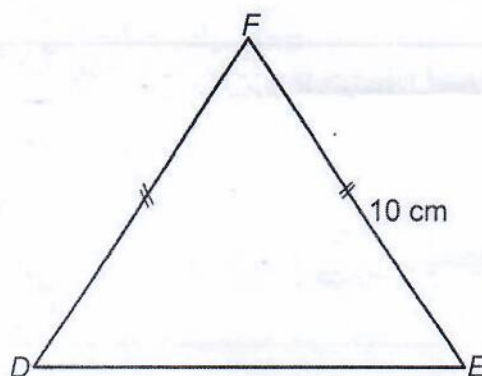
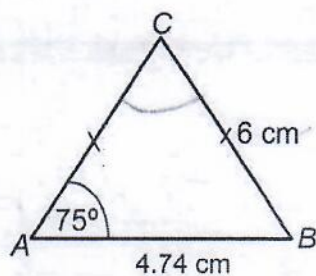
(a) Write down the 5th term of the sequence.

..... (1)

(b) What is the rule, in terms of n , for finding the next term?

..... (2)

The diagram below shows two similar triangles ABC and DEF .
Use the triangles to answer question 16.



16. (a) Calculate the size of angle ACB .

..... (2)

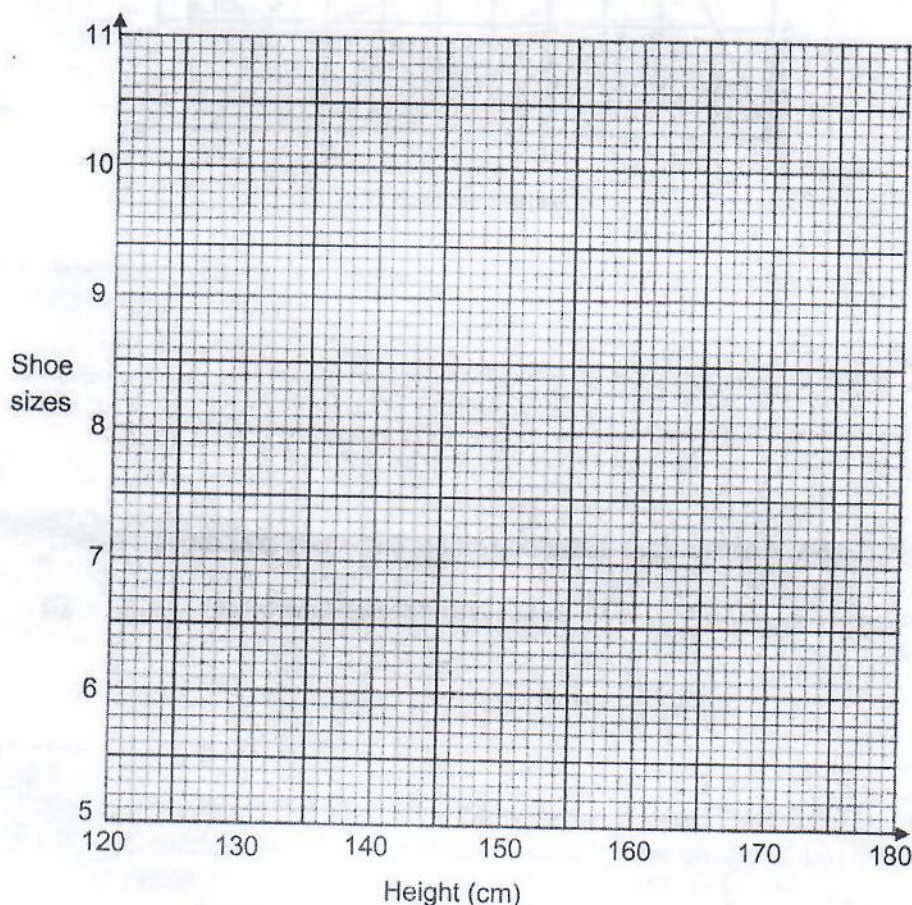
(b) Calculate the length of line segment DE .

..... cm (2)

The table below shows the heights of 8 men in a team and their corresponding shoe sizes. Use it to answer question 17.

Height (cm)	140	145	145	160	160	165	170	175
Shoe size	6	7	6	9	8	9	10	11

17. (a) In the pair of axes below, plot the points for a scatter diagram of the heights and the corresponding shoe sizes of men.

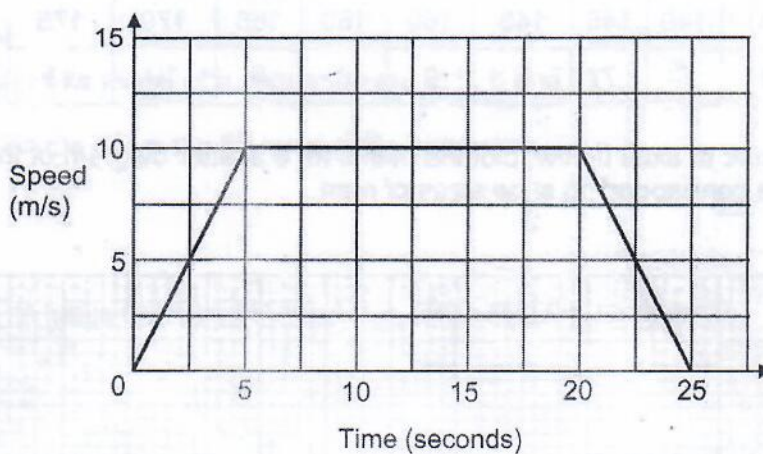


- (b) Draw a line of best fit for the scatter diagram.

- (c) A 9th man who wants to join the team has a shoe size of $10\frac{1}{2}$. Use the line of best fit to estimate the height of the man.

.....cm (1)

The diagram below shows Tapiwa's speed-time graph during a race. Use it to answer question 18.



18. (a) Calculate the acceleration during the first 5 seconds of the race.

..... m/s^2 (2)

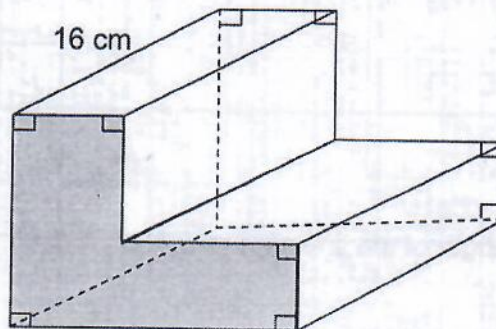
- (b) Show that the total distance ran by Tapiwa is 200 m.

..... m (2)

- (c) Calculate Tapiwa's average speed in m/s.

..... m/s (2)

The diagram below shows a metal piece with a shaded cross-section. Use it to answer question 19.



19. (a) The area of the cross-section is 29 cm^2 . Calculate the volume of the metal piece.

..... cm^3 (2)

- (b) The metal piece is melted and shaped into small cubes of side 6 cm.

- (i) Calculate the volume of one cube.

..... cm^3 (2)

- (ii) Calculate the number of cubes that can be shaped from the metal piece.

..... (3)

20. The coordinates of point A are (0, 7) and those of point B are (8, 0).

- (a) Calculate the coordinates of the midpoint of the line segment joining A and B.

..... (2)

- (b) Calculate the length of the line segment AB.

..... (2)

- (c) What is the equation of the line passing through the points A and B?

..... (2)

The table below shows values of x and the corresponding values of y for the equation $y = x^2 + 3$. Use it to answer question 21.

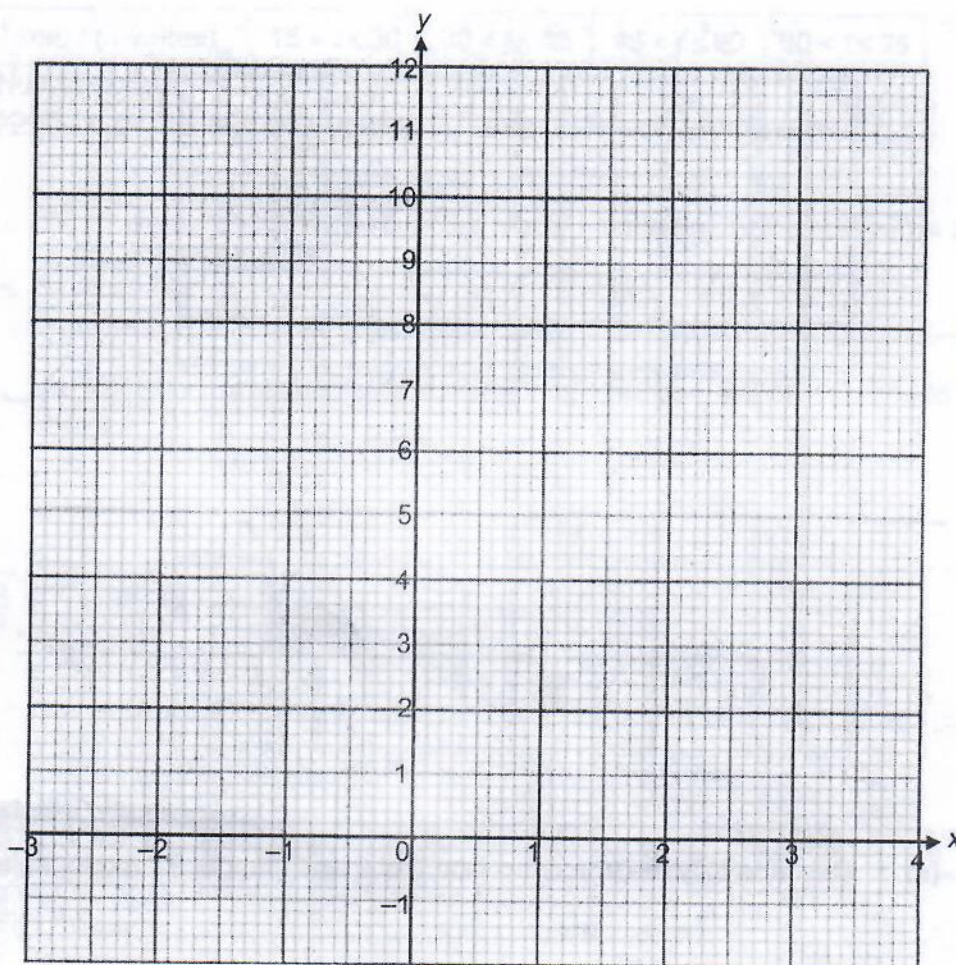
x	-2	-1	-0.5	0	0.5	1	2	3
y	7	4	3.25	3	3.25	4	7	r

21. (a) Calculate the value of r .

..... (1)

- (b) Use the grid below to draw the graph of the equation $y = x^2 + 3$.

(3)



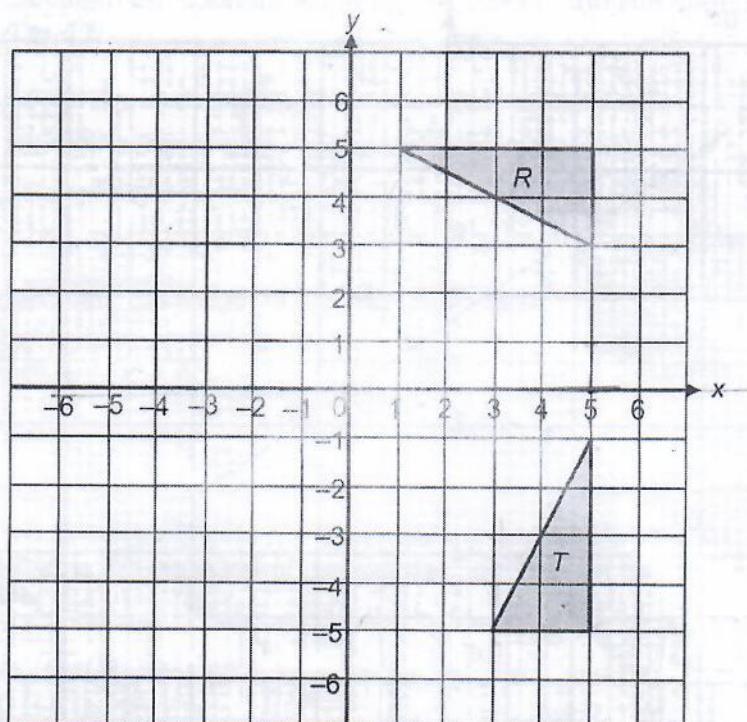
- (c) Draw the graph of the line $y = 5$ on the grid above.

(1)

- (d) Use the graphs to find the values of x that satisfy the equations $y = 5$ and $y = x^2 + 3$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ (2)

Use the diagram below to answer question 22.

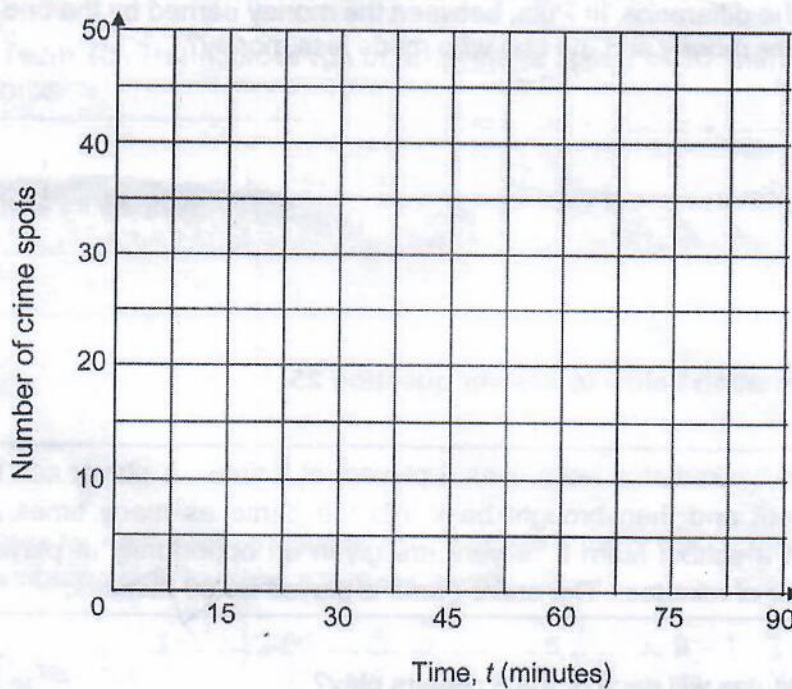


22. (a) Describe fully the transformation that maps triangle R onto triangle T .
..... (3)
- (b) Triangle R reflected along the y -axis. Draw and label its image U . (2)

The table below shows the time taken by the police to arrive at 105 crime spots. Use it to answer question 23.

Time, t (minutes)	$15 < t \leq 30$	$30 < t \leq 45$	$45 < t \leq 60$	$60 < t \leq 75$
Number of crime spots	20	40	35	10

23. (a) Which time interval represents the modal time taken by the police to arrive at the crime spots?
..... (1)
- (b) At how many crime spots did the police take 45 minutes or less time to arrive?
..... (2)



- (c) Draw a histogram on the pair of axes above to represent the information shown in the table. (2)
- (d) On the same pair of axes draw a frequency polygon. (2)

SECTION C

(20 Marks)

Answer all questions in this section.

24. Sello and Mpho answer 5 questions each in a quiz. They each earn P30 for any correct answer and lose 10% of the total amount of money they have accumulated if they answer incorrectly.

Sello answers the 1st, the 3rd, the 4th and the 5th questions correctly but answers the 2nd question incorrectly.

Mpho answers the **first four** questions correctly but answers the **last** question incorrectly.

- (a) Who made more money from answering the questions?

..... (3)

- (b) What is the difference, in Pula, between the money earned by the one who made more money and the one who made less money?

P..... (1)

Use the information below to answer question 25.

In a game of basketball a team uses 5 players at a time. A player can be substituted out and then brought back into the game as many times as possible. In a school team 8 players are given an opportunity of playing equal number of minutes. The entire game is played for 80 minutes.

25. How many minutes will each of the 8 players play?

..... minutes. (3)

26. A teacher rewards the first top 5 students in a test. He gives 5 points to the student who is 5th, 15 points to the 4th student, 45 points to the 3rd student and goes on until he reaches the student who is 1st.

How many points will be awarded to all the five students?

..... (3)

Use the information below to answer question 27.

Teams A and B participate in a 5 000 m race using two different ways that are described below.

Team A: The athletes run **half** of the race at an average speed of 12 km/h and the remainder of the race at an average speed of 6 km/h.

Team B: The athletes run at an average speed of 10 km/h throughout the race.

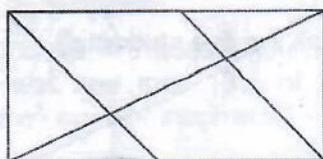
27. Which team is likely to finish the race first?
Show all working.

..... (4)

28. Place four different operations +, −, × or ÷ in each of the spaces between the numbers below to make a true statement.

3 50 5 6 2 = 1 (2)

The diagram below shows a rectangle divided into triangles and quadrilaterals. Use it to answer question 29.



29. (a) How many triangles are there in the diagram?
 (2)
- (b) How many quadrilaterals are there in the diagram?
 (2)



BOTSWANA EXAMINATIONS COUNCIL
JUNIOR CERTIFICATE EXAMINATION

MATHEMATICS

13/2

Paper 2

October/November 2015

Marks: 100

Time: 2 Hours

Candidate's Examination Number:

Centre						Candidate			

INSTRUCTIONS

1. Write your examination number in the space provided above.
2. Answer **ALL** questions.
3. **ALL** answers must be written in the spaces provided.
4. Show all the necessary working.
5. Calculators may be used in this paper.
6. Use the value π in your calculator unless otherwise stated.
7. If the degree of accuracy is **not** specified in the question and the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

FOR EXAMINER'S USE ONLY

Section	Marks Scored
A	
B	
C	
Total Marks	

This question paper contains 17 printed pages.

Section A

(30 Marks)

Answer all questions in this section

1. Calculate $\frac{2}{7}$ of P176.40.

P..... (2)

2. In the formula $F = \frac{mn}{r^2}$, calculate the value of F when $m = 38.7$, $n = 5$ and $r = 4.3$.

..... (2)

3. Solve the equation $3x + 2 = 5$.

 $x =$ (2)

4. What is $3\begin{bmatrix} 5 \\ -2 \end{bmatrix} + \begin{bmatrix} -3 \\ 8 \end{bmatrix}$ when expressed as a single vector?

 $\begin{bmatrix} \\ \end{bmatrix}$ (3)

5. Factorise $15ab - 10b + 5$ completely.

..... (2)

6. Expand $(x + 4)^2$.

..... (2)

7. The temperatures in $^{\circ}\text{C}$ for 8 different places were recorded as follows:

-1, 2, 10, 8, 7, 3, 9, 5.

- (a) What is the median temperature?

..... $^{\circ}\text{C}$ (1)

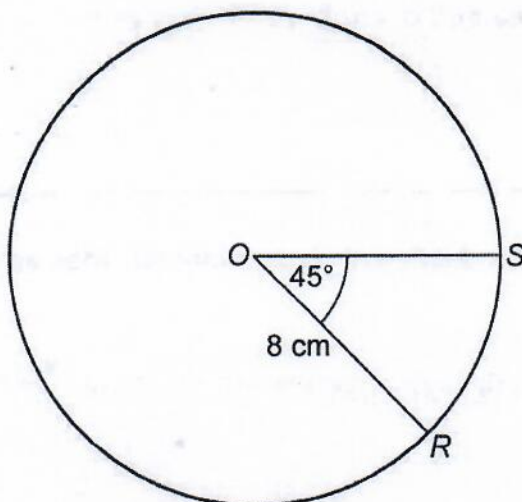
- (b) What is the difference between the highest temperature and the lowest one?

..... $^{\circ}\text{C}$ (1)

8. Factorise $hs + ht + ks + kt$ completely.

..... (3)

9. The diagram below shows a circle with centre O and a radius of 8 cm.



- (a) Calculate the circumference of the circle.
Use π as 3.142.

.....cm (2)

- (b) Calculate the length of the **minor** arc RS .

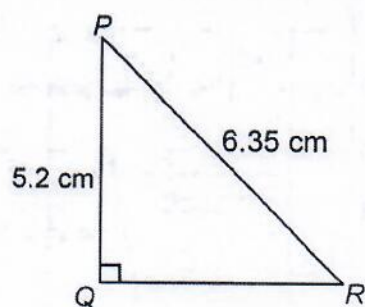
.....cm (2)

-
10. A sequence is given as 3, 7, 11, 15, 19, 21 ...

What is the rule for finding the next term of the sequence?

..... (2)

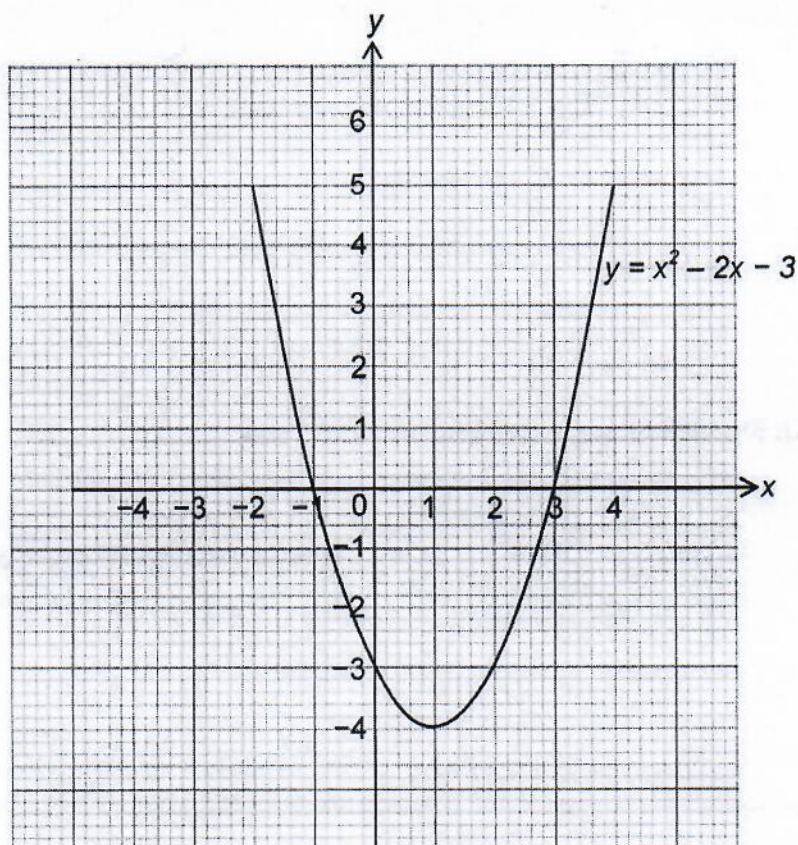
11. A right angled triangle PQR is shown below.



Calculate the size of angle QPR .

.....° (2)

12. The diagram below shows the graph of $y = x^2 - 2x - 3$.



- (a) Draw the graph of the line $y = -3$ (1)
- (b) Solve the equation $x^2 - 2x - 3 = -3$

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ (3)

Answer all questions in this section.

13. The table below shows the price ranges for cars and corresponding percentage discount.

Price (P)	Discount
10 000 – 25 000	6.4%
26 000 – 42 000	5.7%
43 000 – and above	4.8%

- (a) Thero bought a car valued at P57 000.

- (i) What percentage discount did he get?

..... (1)

- (ii) Calculate the total amount of money he had to pay.

..... (3)

- (b) Kago bought two cars valued at P13 750 and P32 500.

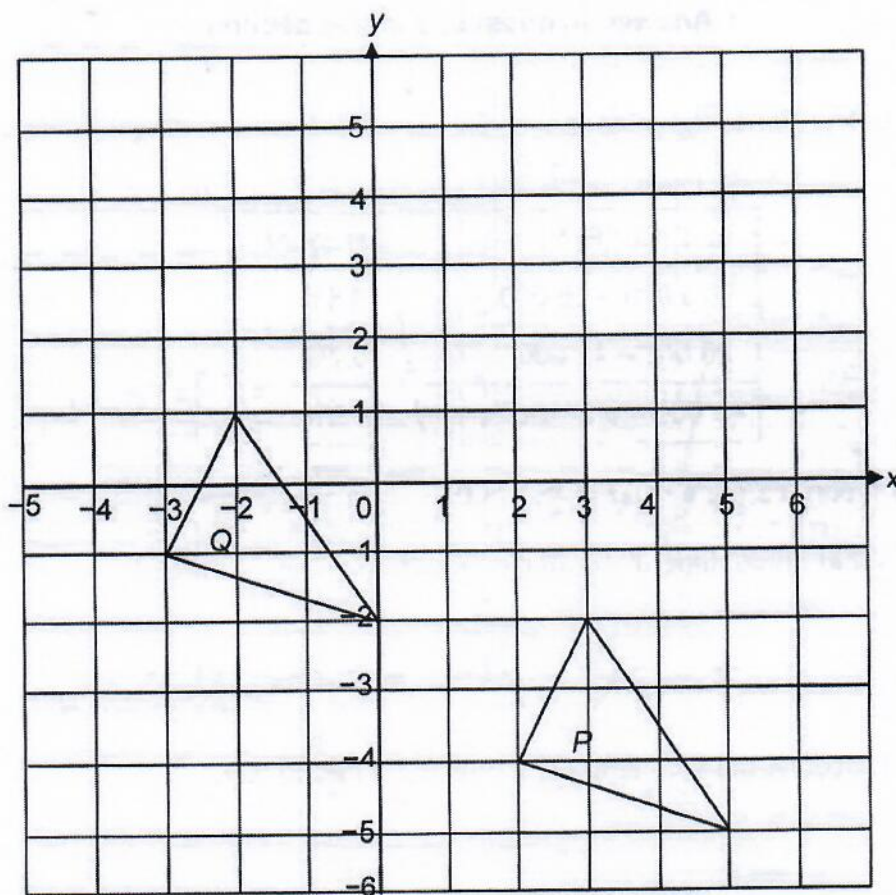
- (i) Calculate the total amount of money he saved from the discount.

P (1)

- (ii) Express the amount of money saved as a percentage.
Give your answer correct to 2 decimal places.

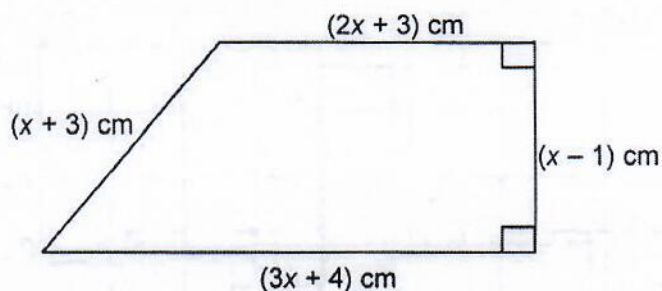
.....% (2)

14. In the diagram below triangle Q is mapped onto triangle P by a single transformation.



- (a) Describe the transformation that maps Q onto P .
..... (2)
- (b) Triangle P is rotated through 90° in an anticlockwise direction about $(1, 0)$. Draw its image and label it R . (3)

15. The diagram below shows a trapezium.



- (a) Write down an expression, in terms of x , for the perimeter of the trapezium.

..... (1)

- (b) The perimeter of the trapezium is 58 cm.

- (i) Form an equation, in terms of x , for the perimeter of the trapezium.

..... (1)

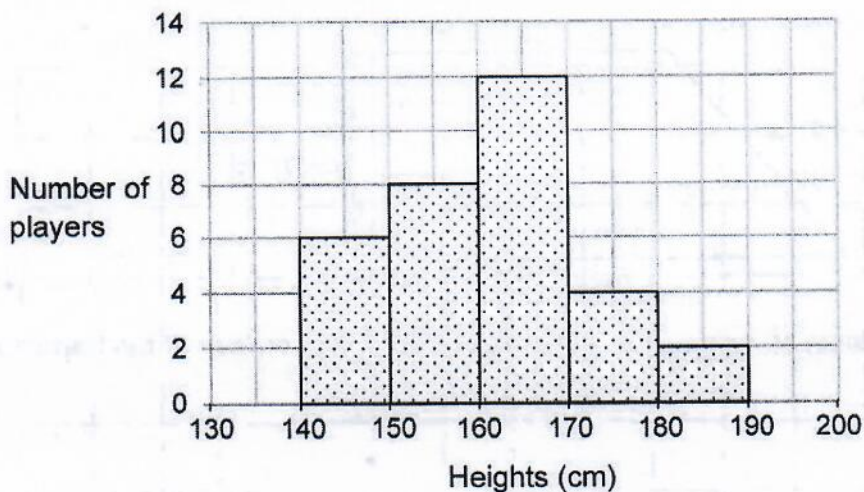
- (ii) Solve the equation in (b) (i).

..... (2)

- (c) Calculate the area of the trapezium.

..... cm^2 (2)

16. The histogram below shows the heights of rugby players.



(a) What is the modal class of the player's heights?

..... cm (1)

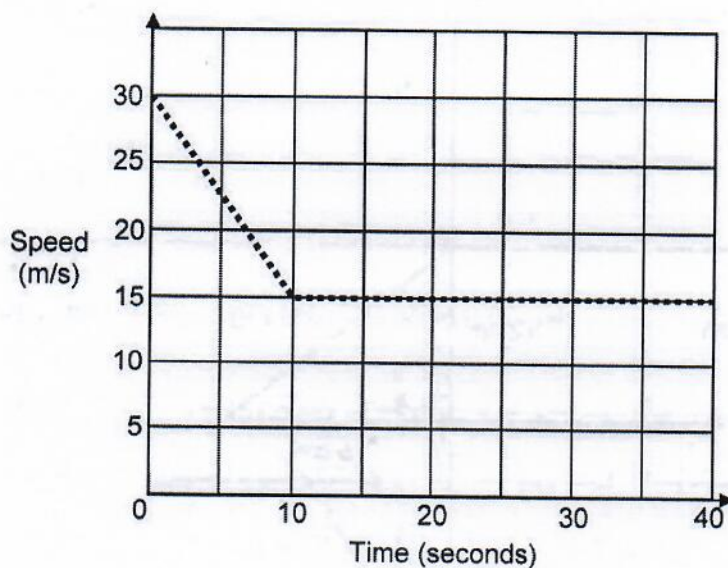
(b) How many players are 160 cm tall or more?

..... (2)

(c) On the histogram above, draw a frequency polygon for the heights of the rugby players.

(2)

17. The speed-time graph of a car over a period of 40 seconds is shown below.



- (a) Calculate the deceleration of the car during the first 10 seconds.

..... m/s^2 (2)

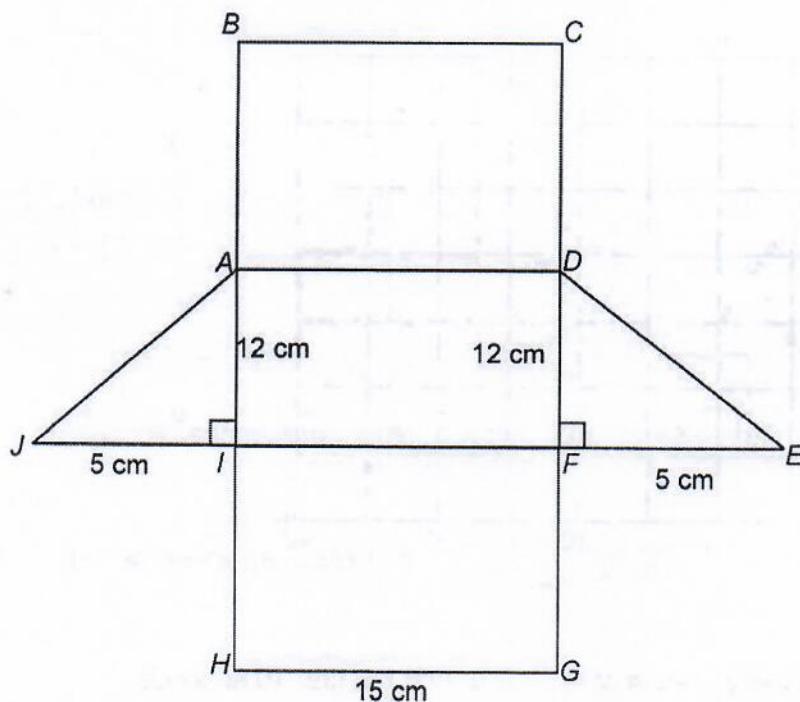
- (b) (i) Calculate the total distance covered by the car in the 40 seconds.

..... m (3)

- (ii) Calculate the average speed of the car.

..... m/s (2)

18. The diagram below shows a sketch of a net.



- (a) Calculate the length of DE .

..... cm (2)

- (b) Calculate the **total** surface area of the net.

..... cm^2 (3)

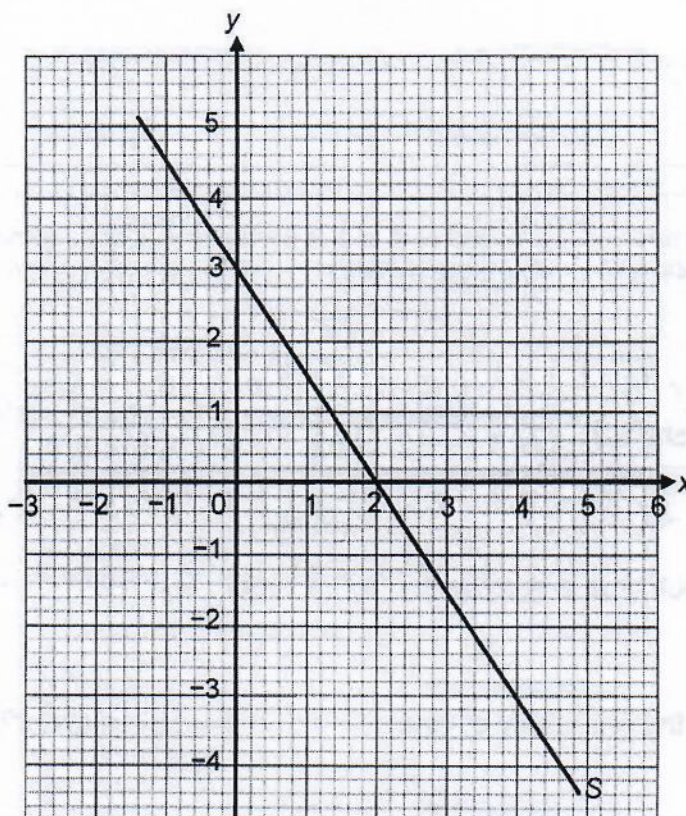
- (c) (i) What is the specific name of the solid that can be made from the net?

..... (1)

- (ii) Calculate the volume of the solid that would be made from the net.

..... cm^3 (2)

19. The graph of a straight line S is shown below.



- (a) (i) Show that its gradient is $-\frac{3}{2}$. (2)

- (ii) What is its equation?

..... (1)

- (b) The coordinates of point T are $(3, 2)$ and those of point U are $(5, 4)$.

(i) What are the coordinates of the midpoint of the line segment TU ?

..... (2)

(ii) Calculate the distance between the points T and U .

..... (2)

20. In an experiment, a die is rolled and at the same time a coin is tossed. The possible outcomes are shown in the table below.

		Die					
		1	2	3	4	5	6
Coin	H	$H, 1$	$H, 2$	$H, 3$	$H, 4$	$H, 5$	$H, 6$
	T	$T, 1$	$T, 2$	$T, 3$	$T, 4$	$T, 5$	r

(a) Which outcome is represented by the letter r ?

..... (1)

(b) What is the probability of obtaining a T and an odd number?

..... (2)

(c) What is the probability of obtaining an outcome in which the die shows a number less than 5?

..... (2)

Answer all questions in this section.

21. In a game, a correct answer scores 7 points and an incorrect one leads to a loss of 3 points. Wazha had a total of 22 points after answering 6 questions.

How many of the questions did he get correct and how many did he get wrong?

Correct (1)

Wrong (1)

22. The diagrams below show a pattern made of triangles and rectangles.

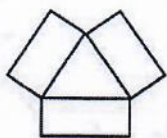


Diagram 1

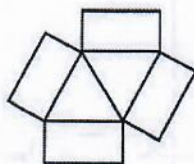


Diagram 2

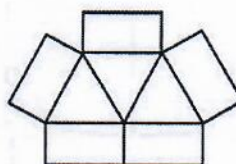


Diagram 3

- (a) In the space below, draw diagram 4.

(2)

- (b) What is the total number of rectangles that would be in diagram 7?

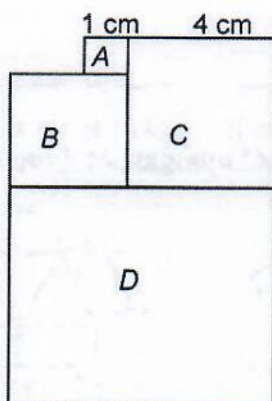
..... (1)

23. A farmer erected 15 poles in a row at equal distances from each other. The distance between two successive poles is 3 metres and the diameter of each pole is 0.085 metres.

Calculate the distance between the 1st pole and the 15th pole.

..... metres (3)

24. The diagram below shows a shape made up of squares A, B, C and D.



Calculate the area of :

- (i) square B
(ii) square D

(i) (2)

(ii) (2)

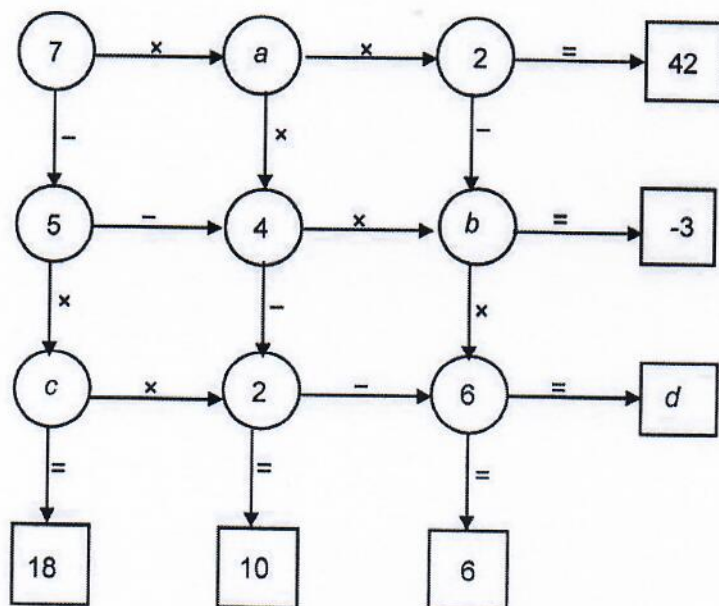
25. In the equation $29 = P + 2^n$, P is a prime number and n is a whole number. Find the values of P and n .

$P = \dots\dots\dots$ and $n = \dots\dots\dots$ (2)

26. The mass of a bull is 325 kg. This is 175 kg more than the mass of a calf. The mass of the calf is 107 kg less than the mass of a cow. Calculate the mass of the cow.

..... kg (2)

27. The diagram below shows mathematical statements. The statements can be read from left to right and from top to bottom.



What are the values of a , b , c and d in the diagram?

a =

b =

c =

d = [4]



BOTSWANA EXAMINATIONS COUNCIL
JUNIOR CERTIFICATE EXAMINATION

MATHEMATICS

13/2

Paper 2

October/November 2012

Marks: 100

Time: 2 Hours

Candidate's Examination Number:

Centre						Candidate			

INSTRUCTIONS

1. Write your examination number in the space provided above.
2. Answer **ALL** questions.
3. **ALL** answers must be written in the spaces provided.
4. Marks will be lost if all necessary working is not shown.
5. Calculators may be used in this paper.
6. Use the value π in your calculator unless otherwise stated.
7. If the degree of accuracy is **not** specified in the question and the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

FOR EXAMINER'S USE ONLY

Section	Marks Scored
A	
B	
C	
Total Marks	

This question paper contains 18 printed pages.

DO NOT TURN THE PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

(30 Marks)

Answer all questions in this section.

1. Write down **all** the prime numbers from the list below;
5, 36, 12, 9, 23, 37.

..... (2)

2. Calculate the value of $3^9 \times 3^{-5}$.

..... (2)

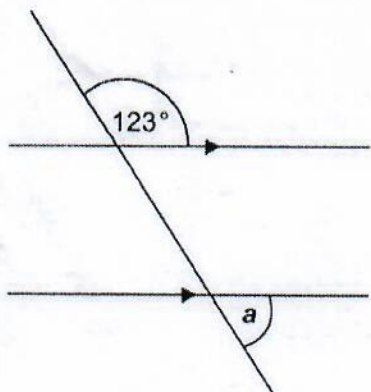
3. Calculate the mean of the numbers 2, 7, 14, 3 and 4.

..... (2)

4. A school bought 560 litres of milk to be used during a family fun day.
The cost of one litre of milk is P11.45.
Calculate the total cost of milk bought for the fun day.

P..... (2)

5. Calculate the size of angle a in the diagram below.



..... (2)

6. The value of a car is P115 000. The value depreciates by 5% every year. Calculate the value of the car after one year.

P..... (2)

7. What is $\begin{pmatrix} 6 \\ 8 \end{pmatrix} - \begin{pmatrix} 6 \\ -7 \end{pmatrix}$ expressed as a single vector?

..... $\begin{pmatrix} \\ \end{pmatrix}$ (2)

8. The coordinates of the point A are $(3, 7)$ and those of the point B are $(5, 2)$. Calculate the length of the line segment AB .

..... (2)

9. Factorise $x^2 - 3x + 2$ completely.

..... (2)

10. Make q the subject of the formula $P = 2q^2 + 3t$.

..... (2)

11. It takes 5 people 4 hours to clean a hall.
How many hours will it take 8 people to clean the hall if they each clean at the same rate?

.....hours (2)

12. A tank had 5 000 litres of water. Some $12\frac{1}{2}\%$ of the water was used for irrigation. How many litres of water were used for irrigation?

..... litres (2)

13. Express $\frac{3}{x} + \frac{2}{x+1}$ as a single fraction.

..... (3)

14. When two coins are tossed, there are four possible outcomes. One of the outcomes is a head from the first coin and a tail from the second coin.

(a) What are the other **three** possible outcomes?

.....
.....
..... (2)

(b) What is the probability that when the two coins are tossed they will show a tail each?

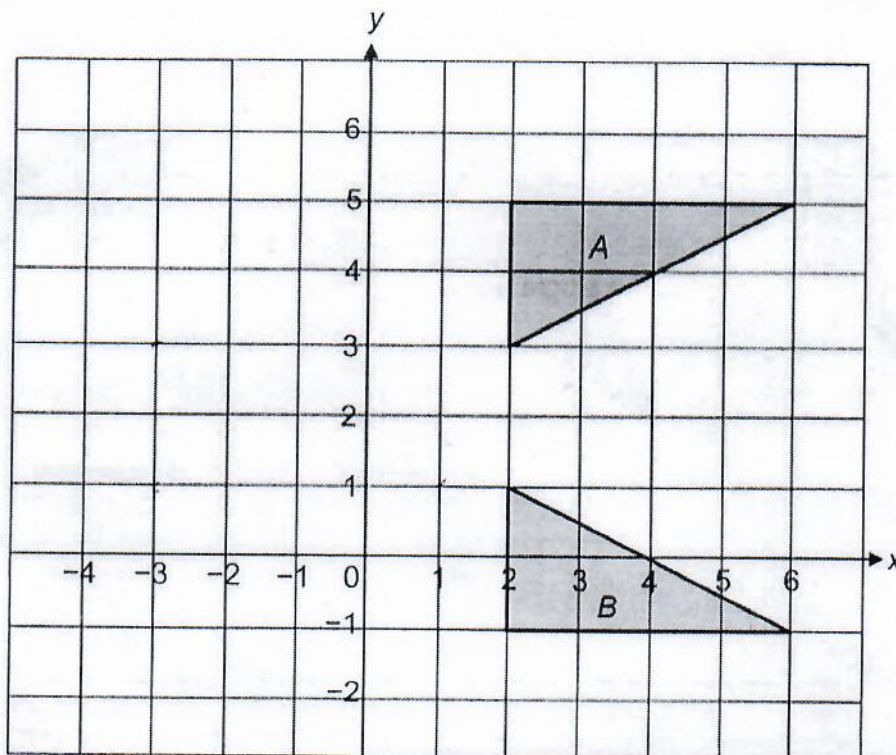
..... (1)

SECTION B

(50 Marks)

Answer all questions in this section.

The diagram below shows triangles *A* and *B*. Use it to answer question 15.



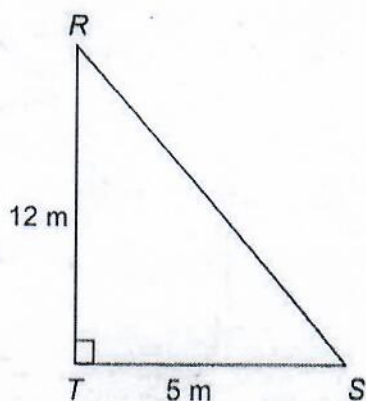
15. (a) Triangle *B* is the image of triangle *A* under a transformation. Describe the transformation fully.

.....
..... (2)

- (b) Triangle *B* is translated onto triangle *C* by vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$. Draw and label triangle *C* on the grid above.

(2)

The diagram below represents a vertical pole, TR , supported by a cable, RS . Use it to answer question 16.



16. (a) Calculate the length of the cable, RS .

.....m (2)

- (b) Calculate the size of the angle RST .

..... (2)

17. (a) Triangle ABC is such that $AB = 8\text{ cm}$, $BC = 6\text{ cm}$ and $AC = 5\text{ cm}$. Construct triangle ABC .

(3)

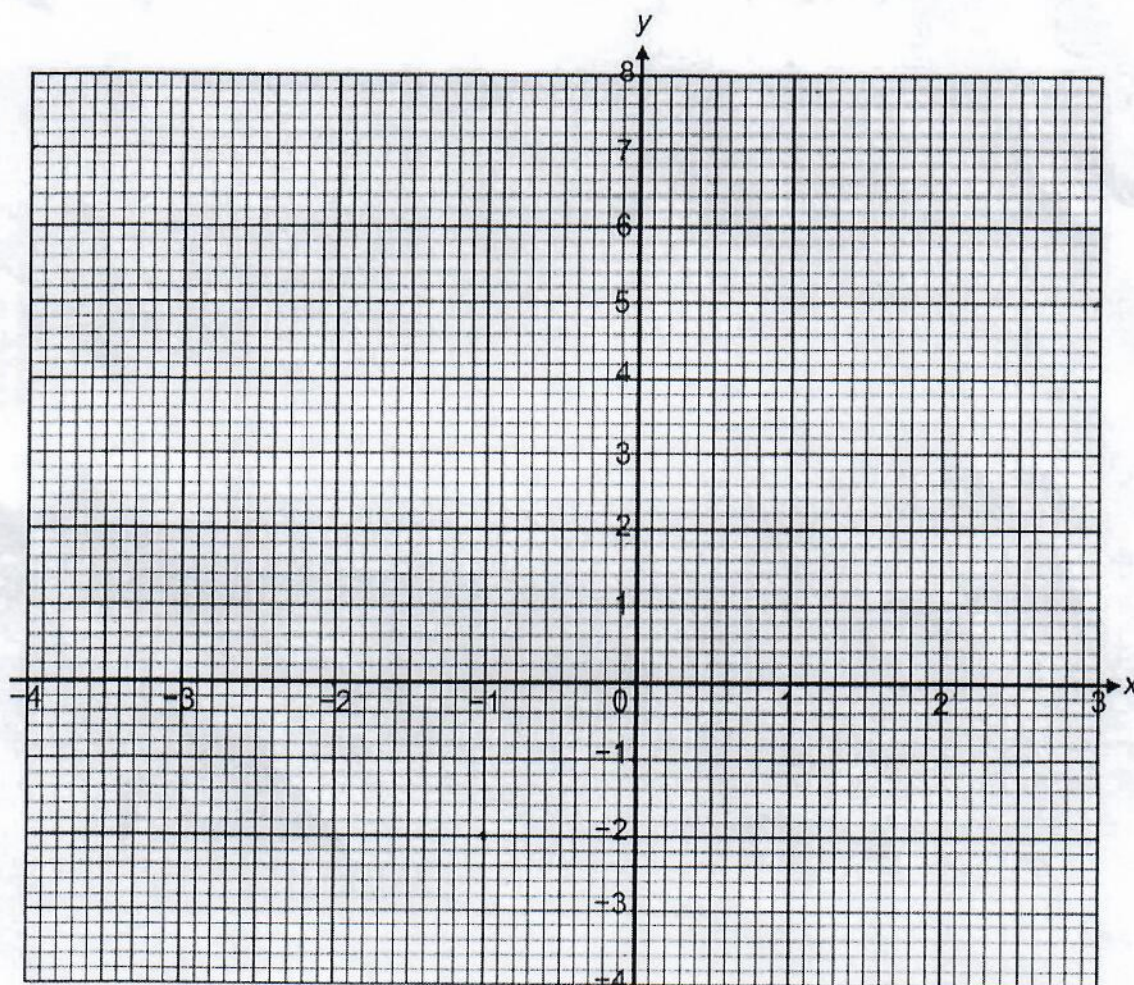
- (b) Construct a line bisector of the side AB .

(1)

18. The table below shows values of x and the corresponding values of y for the equation $y = x^2 + 2x - 1$.

x	-4	-3	-2	-1	0	1	2
y	7	2	-1	-2	-1	2	7

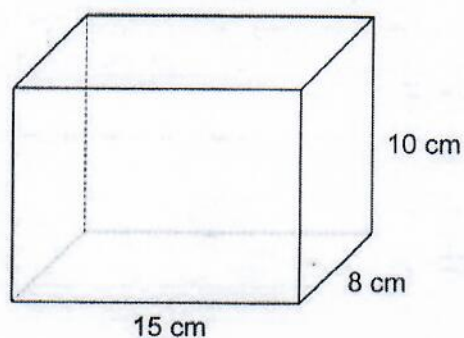
- (a) Use the grid below to plot the points for the graph of $y = x^2 + 2x - 1$. (3)
(b) Join the points with a smooth curve. (1)



- (c) Draw the graph $y = x + 2$ on the grid above. (1)
(d) Use the graphs to estimate the values of x that satisfy both the equations $y = x^2 + 2x - 1$ and $y = x + 2$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ (2)

The diagram below shows a closed box in the form of a cuboid.
Use it to answer question 19.



19. (a) Calculate the total area of all the faces of the box.

.....cm² (3)

- (b) In the space below sketch the net of the box shown above.

(2)

Use the information below to answer question 20.

Neo drove from Gaborone to Francistown starting off at 10 00. She travelled a distance of 200 km for 2 hours to Mahalapye. She rested for 30 minutes in Mahalapye. It then took her a further 3 hours to arrive in Francistown, which is 240 km from Mahalapye.

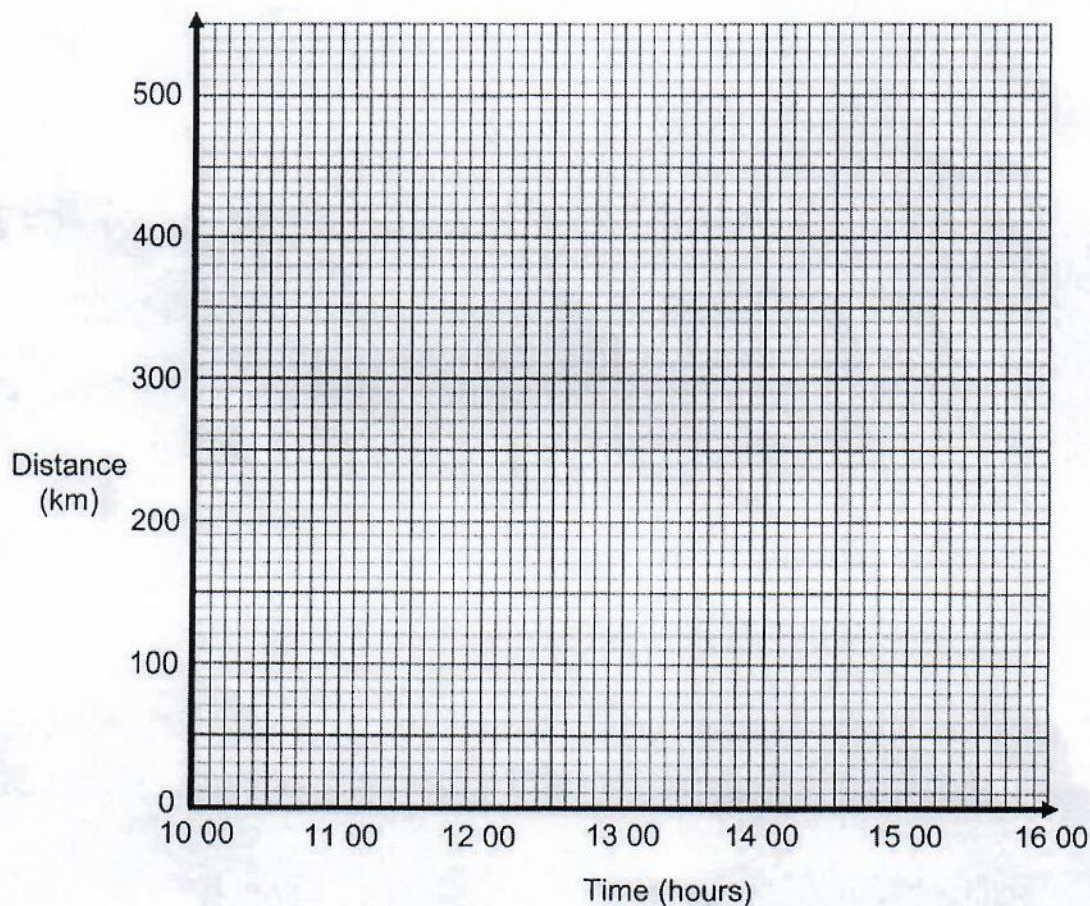
20. (a) At what time did she arrive in Francistown?

..... (1)

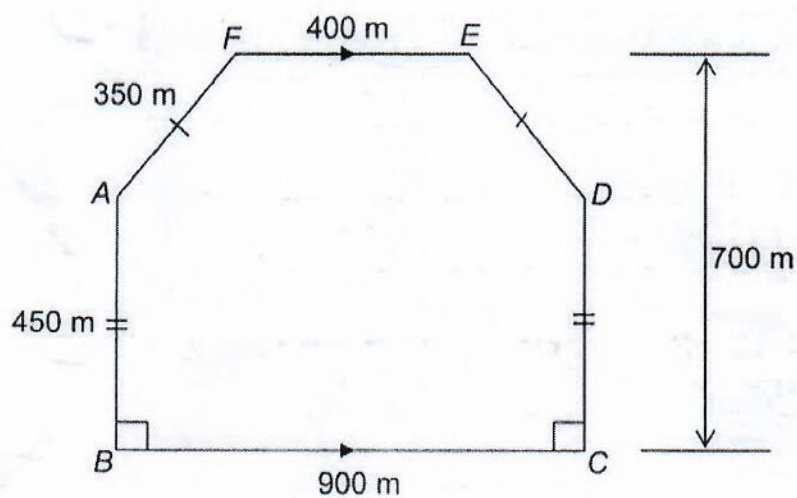
- (b) Calculate the average speed of the car from Mahalapye to Francistown.

.....km/h (2)

- (c) On the pair of axes below, draw a distance-time graph to show Neo's journey from Gaborone to Francistown. (3)



The diagram below represents a plan of a field **ABCDEF**.
Use it to answer question 21.



21. (a) Calculate the perimeter of the field.

.....m (2)

- (b) Calculate the area of the field.

.....m² (3)

The frequency table below shows the lengths of 50 different leaves.
Use it to answer question 22.

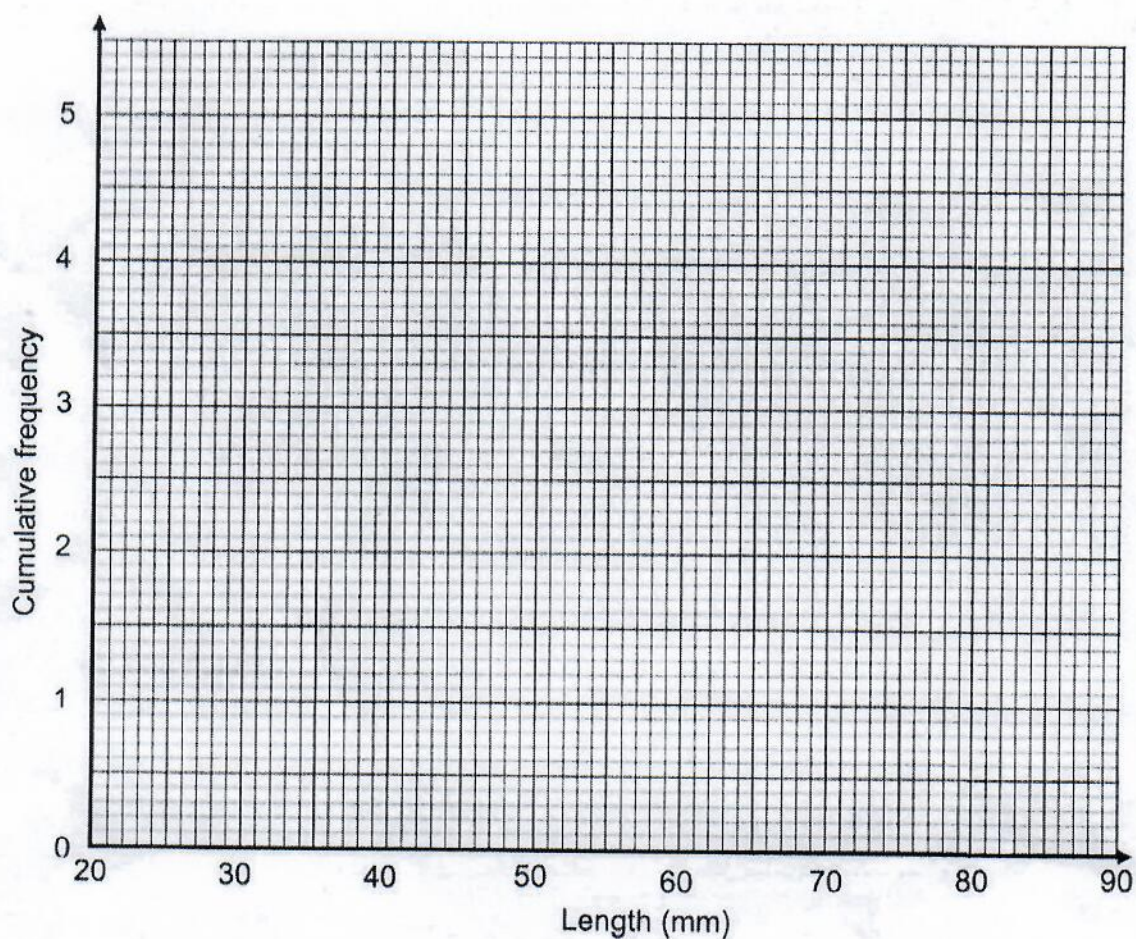
Length (mm) of leaf	Frequency
$20 \leq l < 30$	2
$30 \leq l < 40$	5
$40 \leq l < 50$	8
$50 \leq l < 60$	14
$60 \leq l < 70$	14
$70 \leq l < 80$	5
$80 \leq l < 90$	2
Total	50

22. (a) Complete the cumulative frequency table below for the number of leaves.

Length (mm) of leaf	20	30	40	50	60	70	80	90
Cumulative Frequency	0	2	7	15	29			

(2)

- (b) Plot the points from the cumulative frequency table on the grid provided. (2)



- (c) Join the plotted points on the grid with a smooth curve. (1)
- (d) Use the curve to estimate the median length of the leaves.

..... (1)

- (e) Use the curve to estimate the number of leaves with a length of 45 mm or less.

..... (1)

23. A bundle of spinach costs Pq .

- (a) Express, in terms of q , the cost of 8 bundles of spinach.

P..... (1)

- (b) A bundle of rape costs Pr .

Express, in terms of r , the cost of 14 bundles of rape.

P..... (1)

- (c) The total cost of 8 bundles of spinach and 14 bundles of rape is P58.
Form an equation for the total cost of the bundles of spinach and rape, and
show that it reduces to $4q + 7r = 29$.

..... (2)

- (d) Another equation that can represent the total cost of the bundles of spinach
and the bundles of rape is $3q + 5r = 21$.

Solve the equations in (c) and in (d) simultaneously.

$q = \dots\dots\dots$, $r = \dots\dots\dots$ (3)

- (e) What is the cost of 3 bundles of spinach?

P..... (1)

SECTION C

(20 Marks)

Answer all the questions in this section.

Use the information below to answer question 24.

Goaba bought some oranges. She gave Thobo half of the oranges. Then she gave Lebo half of the number of oranges that were left after giving to Thobo. Goaba was left with only 4 oranges at the end.

24. How many oranges did Goaba buy?

..... (3)

25. Fill in the square below with numbers 1, 2, 3, 4 and 5 such that each number appears only once in each row and each column.

1	3			4
	2		4	1
2		3		5
5	1	4	2	3
		1		2

(3)

26. Study the pattern of numbers below and fill in the blank spaces.

$$2 = 2$$

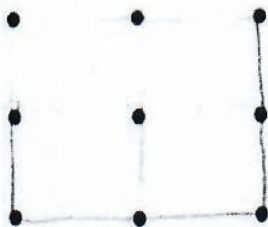
$$2 + 3 = 5$$

$$2 + 3 + 5 = 10$$

$$2 + 3 + 5 + 7 = 17$$

$$2 + 3 + 5 + 7 + \dots = \dots \quad (2)$$

27. (a) Join **all** the 9 dots below using straight lines to make a closed shape. The straight lines should **not** pass through any dot twice.



(2)

(b) What is the name of the polygon formed by joining the dots?

..... (1)

28. What are the values of **a** and **b** in the pattern of numbers below?

$$123 \times 13 \times 11 \times 7 = 123\ 123$$

$$157 \times 13 \times 11 \times 7 = 157\ 157$$

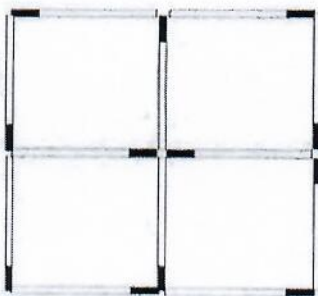
$$a \times 13 \times 11 \times 7 = 208\ 208$$

$$527 \times 13 \times 11 \times 7 = b$$

$$a = \dots, b = \dots \quad (2)$$

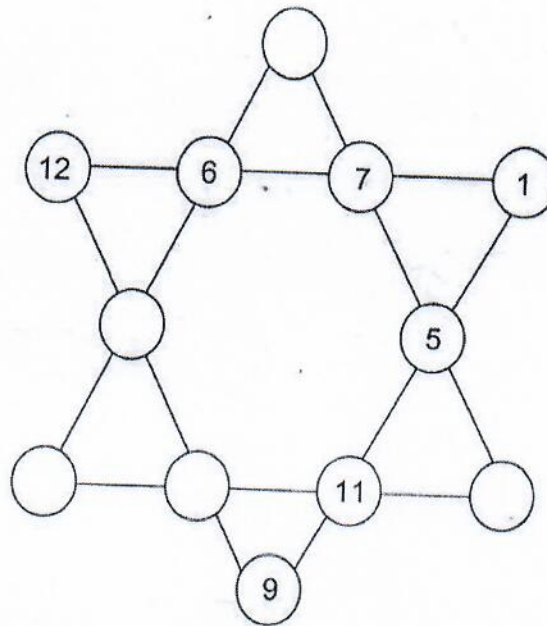
$$\begin{array}{r} 527 \\ \times 13 \\ \hline 15710 \\ + 6850 \\ \hline 68510 \end{array}$$

The figure below shows 12 matchsticks arranged to form squares.
Use it to answer question 29.



29. (a) How many squares have been formed?
..... (1)
- (b) What is the **smallest** number of matchsticks that can be removed such that there is only one square remaining?
..... (1)
- (c) The 12 matchsticks can be arranged to form a composite shape with all the sides equal. Part of the composite shape is a square.
Draw the composite shape in the space below. (2)

30. Fill in the numbers 2, 3, 4, 8 and 10 in the circles so that the sum of the numbers along each of the six straight lines is 26. Each number should be used only once.



(3)



BOTSWANA EXAMINATIONS COUNCIL
JUNIOR CERTIFICATE EXAMINATION

MATHEMATICS

13/2

Paper 2

October/November 2011

Marks: 100

Time: 2 Hours

Candidate's Examination Number:

Centre			Candidate		

INSTRUCTIONS

1. Write your examination number in the space provided above.
2. Answer ALL questions.
3. ALL answers must be written in the spaces provided.
4. Marks will be lost if all necessary working is not shown.
5. Calculators may be used in this paper.
6. Use the value π in your calculator unless otherwise stated.
7. If the degree of accuracy is not specified in the question and the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

FOR EXAMINER'S USE ONLY

Section	Marks Scored
A	
B	
C	
Total Marks	

This question paper contains 20 printed pages.

DO NOT TURN THE PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

(30 Marks)

Answer all questions in this section.

1. Write down the next two terms in the sequence

14, 11, 8, 5,...

(2)

2. Solve the equation $3x - 11 = 5 - x$.

(2)

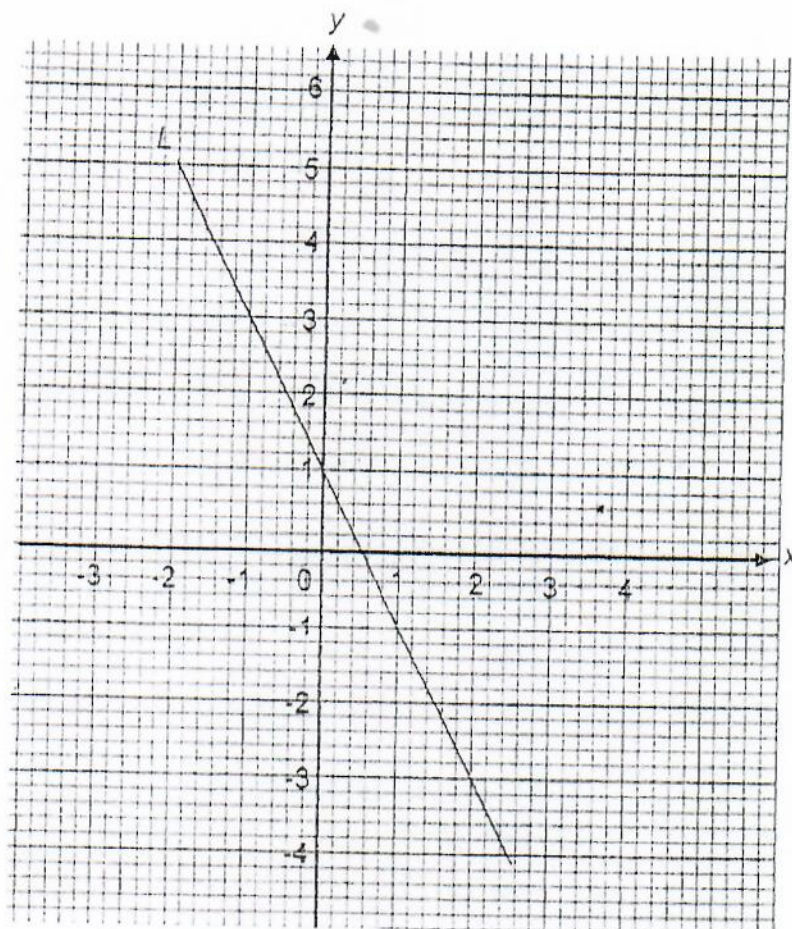
3. The size of an exterior angle of a regular polygon is 40° .
Calculate the number of sides of the polygon.

.....sides (2)

4. In a group of 36 workers, 19 are plumbers and the rest are builders.
A worker is chosen at random from the group.
What is the probability that the worker chosen is a builder?

(2)

5. Find the equation of the line L shown in the grid below.



..... (2)

6. In the formula $5mn^2 = 90$, m and n are prime numbers.
What are the values of

(a) m ,

(b) n ?

$m =$ (1)

$n =$ (1)

7. Solve the simultaneous equations;-

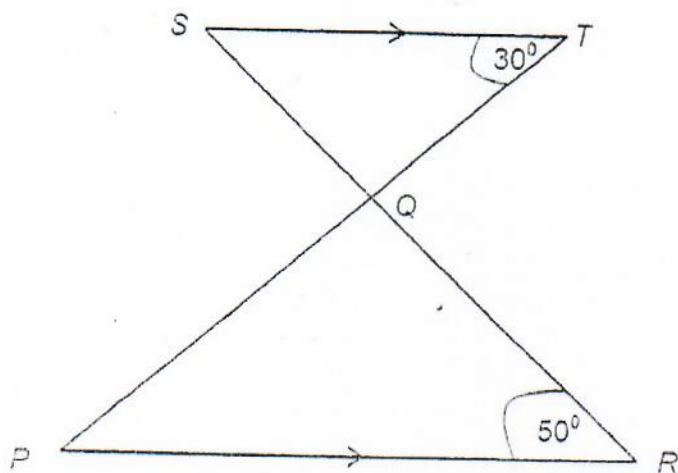
$$\begin{aligned} 3x + y &= 9 \\ 2x - y &= 1. \end{aligned}$$

$$x = \dots\dots\dots y = \dots\dots\dots (3)$$

8. Factorise the expression $x^2 + 7x + 12$ completely.

$$\dots\dots\dots (2)$$

The diagram below shows triangles QST and QRP . The sides ST and PR are parallel. Use the diagram to answer question 9.



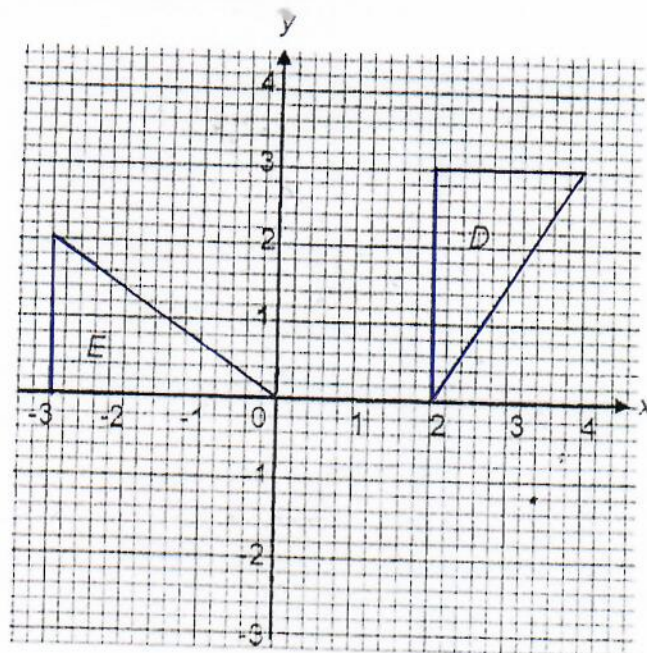
9. (a) State, with a reason, the size of angle TPR .

.....
 (2)

- (b) Calculate the size of angle SQT .

..... (2)

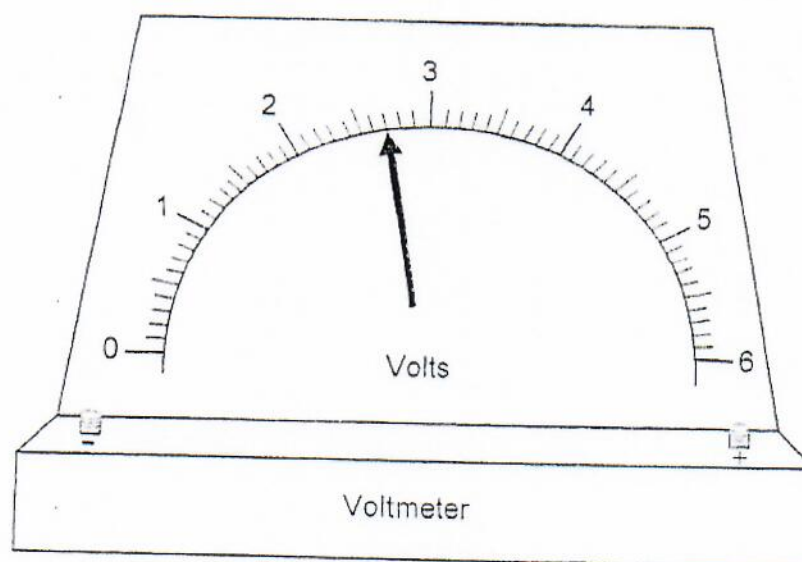
In the diagram below, triangle E is the image of triangle D under a transformation. Use the diagram to answer question 10.



10. Describe fully, the transformation that maps triangle D onto triangle E .

..... (3)

11. What is the reading shown on the voltmeter below?



.....Volts (1)

The table below shows the number of books borrowed by a group of 40 students from a library. Use it to answer question 12.

Number of books	1	2	3	4	5
Number of students	5	11	10	7	7

12. (a) What is the modal number of books borrowed?

..... (1)

- (b) Calculate the range of the number of books borrowed.

..... (1)

- (c) Calculate the mean number of books borrowed.

..... (3)

SECTION B

(50 Marks)

Answer all questions in this section.

13. A customer was given a discount on a laptop that costs P16 000. The customer paid only P14 150 for the laptop.

(a) Calculate the discount?

P..... (1)

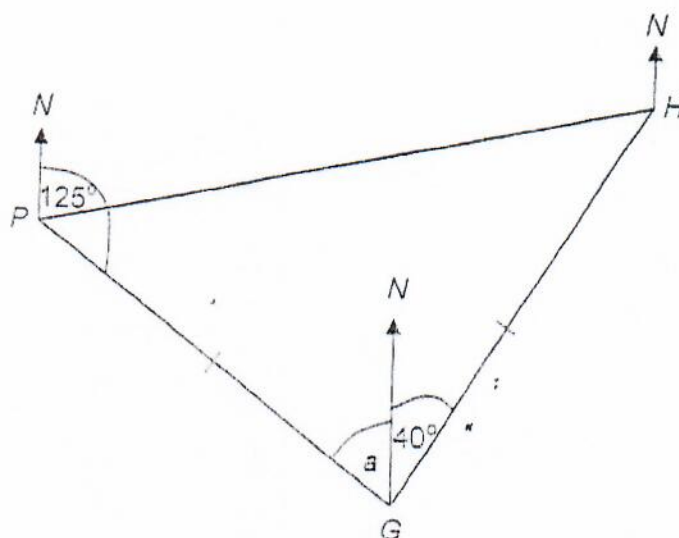
(b) Calculate the percentage discount.

.....% (2)

- (c) At some later stage the customer returned the laptop because she did not like it. She was paid back her money but 5% of the money was retained as handling charges. How much money did the customer receive?

..... (2)

The diagram below shows the positions of points G , H and P where $PG = GH$. Use it to answer question 14.



14. (a) Calculate the size of angle a .

..... (1)

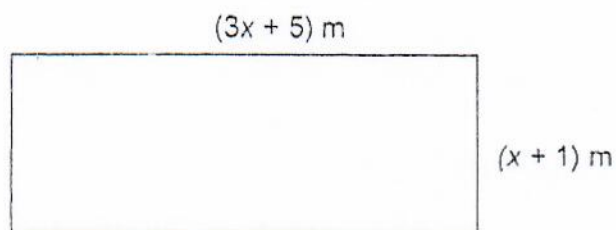
- (b) Calculate the bearing of G from H .

..... (2)

- (c) Calculate the size of angle GHP .

..... (2)

The diagram below represents a rectangular plot of length $(3x + 5)$ m and width $(x + 1)$ m. Use it to answer question 15.



15. (a) Express the perimeter of the plot in terms of x .

.....m (1)

- (b) The perimeter of the plot is 36 m.

- (i) Form an equation, in terms of x , for the perimeter of the plot.

..... (1)

- (ii) Solve the equation in b (i).

..... (1)

- (c) Calculate the area of the plot.

..... m^2 (2)

- (d) The plot is divided into parking spaces. Each parking space is 2.5 m wide and is marked along the **length** of the plot. Calculate the **maximum** number of parking spaces that can be marked on the plot.

..... (2)

The table below shows charges for radio advertising per 30 seconds for day and night times. Use it to answer question 16.

	Charge per 30 seconds
Day time	P360
Night time	P250

16. (a) (i) Lodi Hair Salon paid for 2 advertisements that lasted 30 seconds each during day time.
Calculate the total charge for the advertisements.

P..... (1)

- (ii) The hair salon also paid for 1 advertisement that lasted 2 minutes during night time.
Calculate the total charge for the advertisement.

P..... (2)

- (iii) Calculate the total amount of money that was paid by the hair salon for all the 3 advertisements.

P..... (1)

- (b) Mosu Milling Company paid a total amount of P3 240 for different advertisements to be run during the day time. Each advertisement was to last 45 seconds.

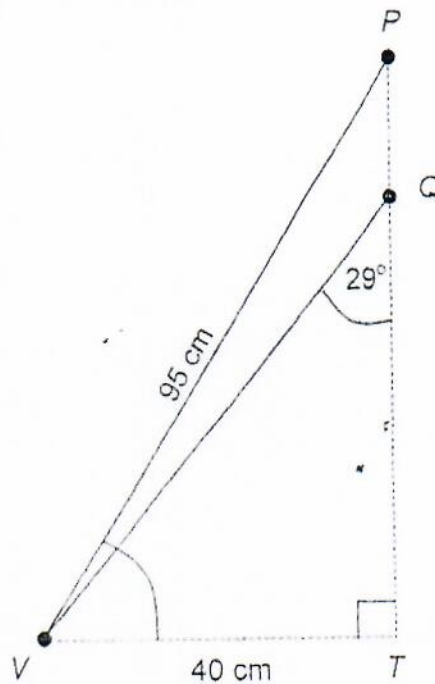
- (i) Calculate the total number of advertisements to be run for the milling company.

..... (2)

- (ii) There should be a 15 minute break between any two advertisements.
If the first advertisement starts at 06 40, at what time will the 4th one end?

..... (2)

The diagram below shows the points P , Q , T and V . The points P , Q and T are in a straight line. Use the diagram to answer question 17.



17. (a) Calculate the length of PT .

.....cm (2)

- (b) Calculate the length of QT .

.....cm (2)

- (c) Calculate the size of angle PVT , correct to 1 decimal place.

..... (2)

- (d) Calculate the length of PQ .

.....cm (1)

18. A customer buys 1 500 bricks at P6.35 each.

(a) Calculate the total cost of the bricks.

P..... (1)

(b) The customer pays 12% of the cost of the bricks as VAT.

(i) Calculate the amount of money paid as VAT.

P..... (2)

(ii) Calculate the total amount of money paid for the bricks.

P..... (1)

(c) It was found that 250 of the bricks bought were damaged. The supplier had enough bricks to replace only $\frac{1}{5}$ of the damaged.

(i) Calculate the number of bricks which the supplier could replace.

..... (1)

(iii) The customer received cash back for the damaged bricks that could not be replaced.

Calculate the amount of money that was paid back to the customer.

..... (2)

The table below shows the grades obtained by 120 students in a Mathematics test. Use it to answer question 19.

Grade	Number of students
A	25
B	r
C	45
D	15
Total	120

19. (a) Calculate the value of r .

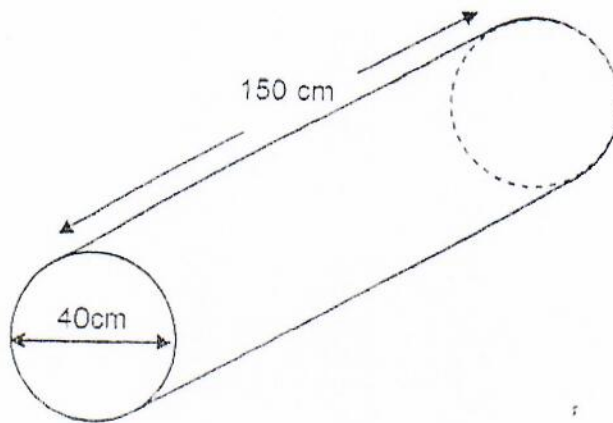
$r = \dots\dots\dots$ (2)

- (b) A pie chart showing the grades of the students is to be drawn.

Calculate the size of the sector angle for students who obtained grade C.

$\dots\dots\dots$ (2)

The diagram below shows a cylindrical pillar of diameter 40 cm and length 150 cm. Use it to answer question 20.



20. (a) What is the name of the shape of the cross-section of the pillar?

..... (1)

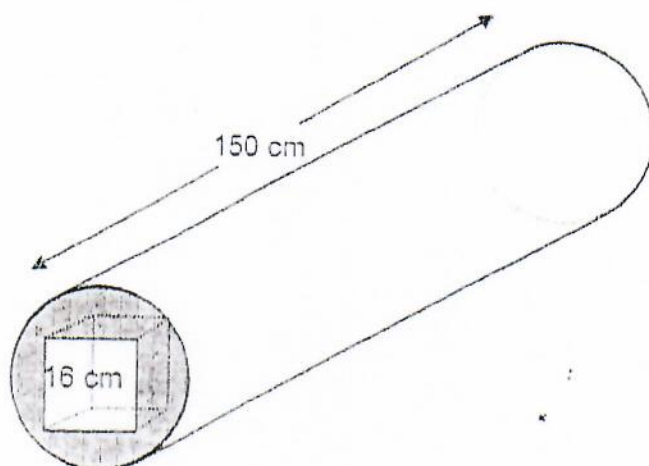
- (b) Calculate the area of the cross-section.

.....cm² (2)

- (c) Calculate the volume of the pillar.

.....cm³ (2)

A portion in the form of a cube of length 16 cm is removed from the end of the pillar as shown in the diagram below.



- (d) Calculate the volume of the portion that is removed.

.....cm³ (2)

SECTION C

(20 Marks)

Answer all questions in this section.

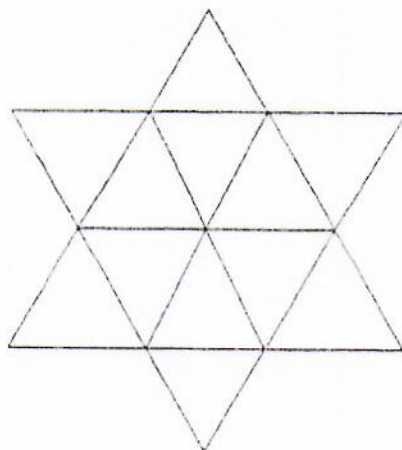
Use the information below to answer question 21.

A car has 5 wheels, 4 of which are fitted and 1 is spare. All the 5 wheels are used on rotational basis throughout a journey of 16 500 km. Each wheel travels the same number of kilometres.

21. Calculate the number of kilometres that each wheel has to travel.

..... (3)

22. What is the total number of triangles shown in the diagram below?



..... (3)

Use the information below to answer question 23.

Thapelo, Kago and Mothusi weighed themselves in pairs and obtained the following results:

Kago and Thapelo: 110 kg
Thapelo and Mothusi: 107 kg
Mothusi and Kago: 121 kg.

23. Calculate the weight of each one of them.

Mothusi = kg

Kago = kg

Thapelo = kg (4)

The diagram below shows sticks arranged such that they cross each other once only. Use the diagram to answer question 24.

Number of sticks	2	3	4	5	6
Arrangement of sticks					
Number of intersections	1	3	6	f	g

24. (a) What are the values of f and g ?

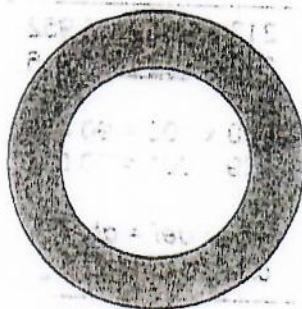
$f =$ (1)

$g =$ (1)

- (b) The number of intersections produces a class of numbers.
What is the name of the class of numbers produced?

..... (1)

The diagram below shows a ring. Use it to answer question 25.



25. A single straight line can be used to divide the shaded region into 2 parts. If three straight lines are used to divide the ring, what is the maximum number of shaded parts that the ring will be divided into?

(2)

Use the information below to answer question 26.

A group of 41 students were given numbers from 1 to 41. The students were arranged in a circle in order of their numbers. They were to remain in their positions until they were ordered to leave the group. Every third student starting from number 1 left the group until only two students remained.

26. What were the numbers of the two students who remained?

(2)

Study the pattern shown below. Use it to answer question 27.

$$\begin{array}{l} 312 \times 221 = 68\,952 \\ 213 \times 122 = 25\,986 \\ \\ 900 \times 100 = 90\,000 \\ 009 \times 001 = 00\,009 \\ \\ abc \times def = gh\,ijk \\ cba \times fed = kj\,ihg \end{array}$$

27. What are the values of p , q and r in the equations below?

$$891 \times 101 = p$$

$$q \times 101 = r$$

$$p = \dots\dots\dots (1)$$

$$q = \dots\dots\dots (1)$$

$$r = \dots\dots\dots (1)$$